

Physical Chemistry (I)

Lecture 19

Special Lecture:

Statistical Mechanics of Proteins



Phase Transitions

Ordered phase

energetically
favored

Interactions



Disordered phase

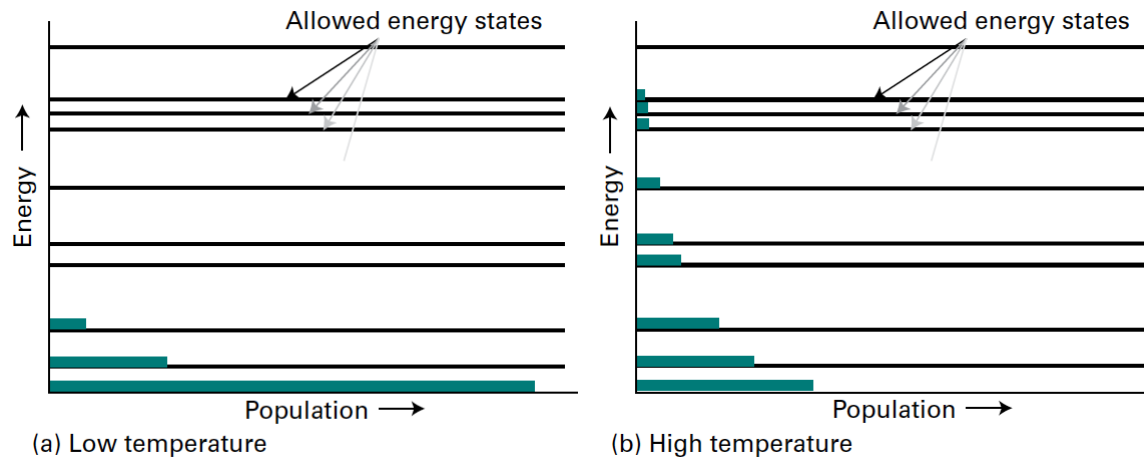
entropically
favored

$$S = k_B \ln W$$

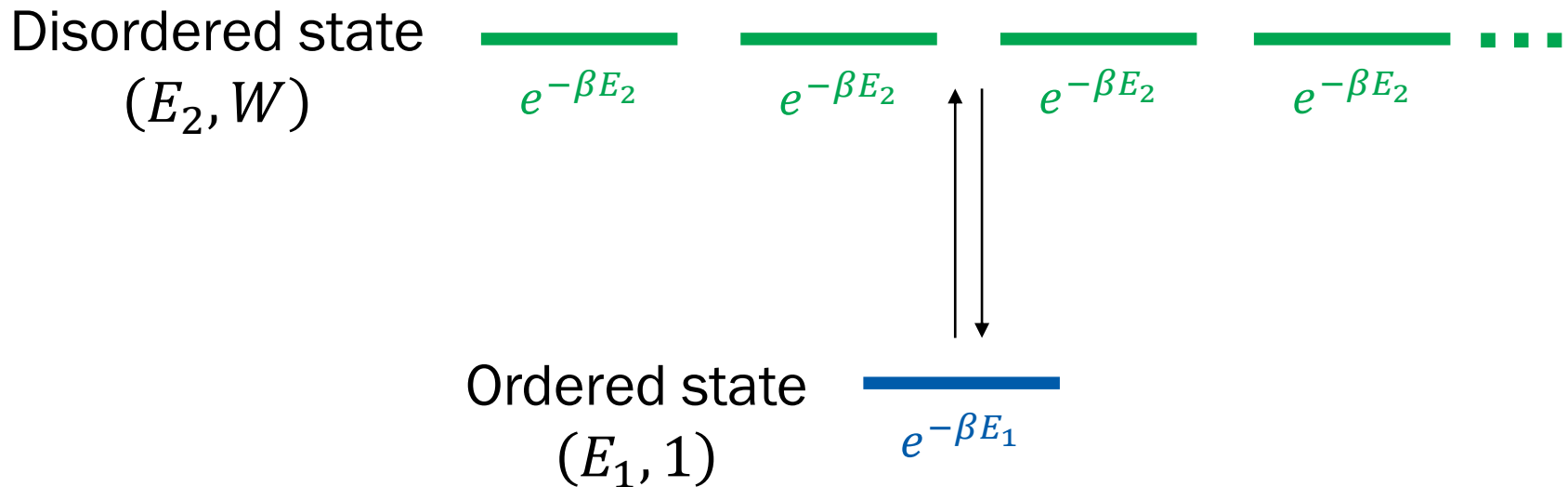
Boltzmann Distribution

$$\text{Probability} \propto e^{-E/k_{\text{B}}T} = e^{-\beta E}$$

(at thermal equilibrium)



Two-State Model

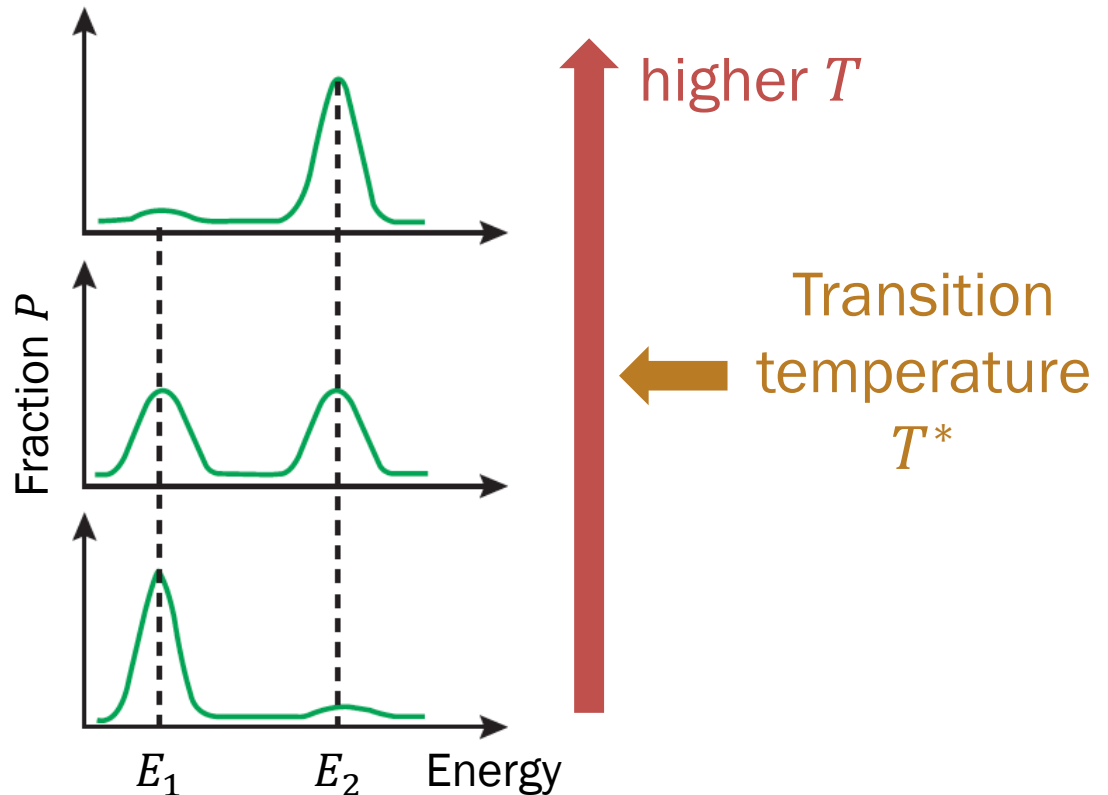


$$P_1 = \frac{e^{-\beta E_1}}{e^{-\beta E_1} + W e^{-\beta E_2}}$$

Phase Transition

$$P_1 = \frac{e^{-\beta E_1}}{e^{-\beta E_1} + W e^{-\beta E_2}}$$

$$P_2 = \frac{W e^{-\beta E_2}}{e^{-\beta E_1} + W e^{-\beta E_2}}$$



Transition Temperature

$$P_1 = \frac{e^{-\beta E_1}}{e^{-\beta E_1} + W e^{-\beta E_2}}, \quad P_2 = \frac{W e^{-\beta E_2}}{e^{-\beta E_1} + W e^{-\beta E_2}}$$

At the transition temperature, $P_1 = P_2$

$$e^{-\beta E_1} = W e^{-\beta E_2} = e^{-\beta(E_2 - T^* S_2)}$$

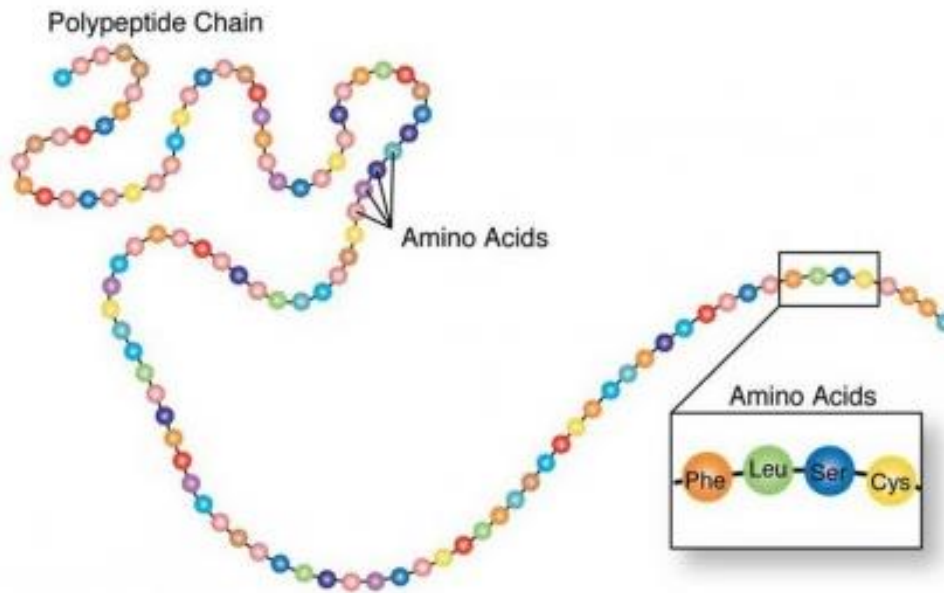
Hence,

$$E_1 = E_2 - T^* S_2$$
$$T^* = \frac{E_2 - E_1}{S_2} = \frac{E_2 - E_1}{k_B \ln W}$$

Now Let's Talk About **Proteins** ...



Proteins are Polymer Chains



Proteins are Polymer Chains

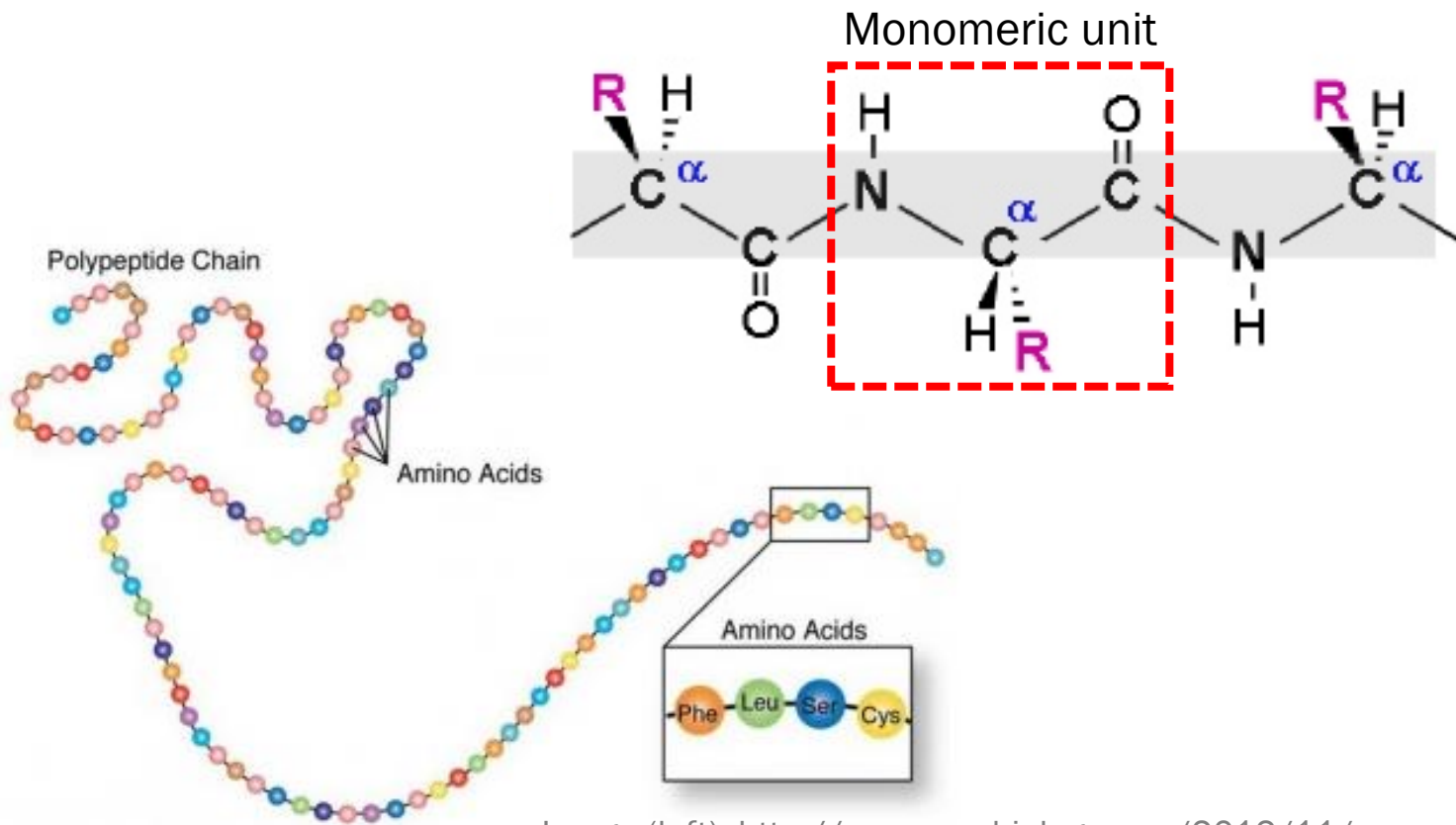


Image (left): <http://www.mcqbiology.com/2012/11/mcq-on-amino-acids.html>

Image (right): <https://www.slideshare.net/MUBOSScz/14-proteins>



Image (right): <https://www.slideshare.net/MUBOSScz/14-proteins>

Proteins are Polymer Chains

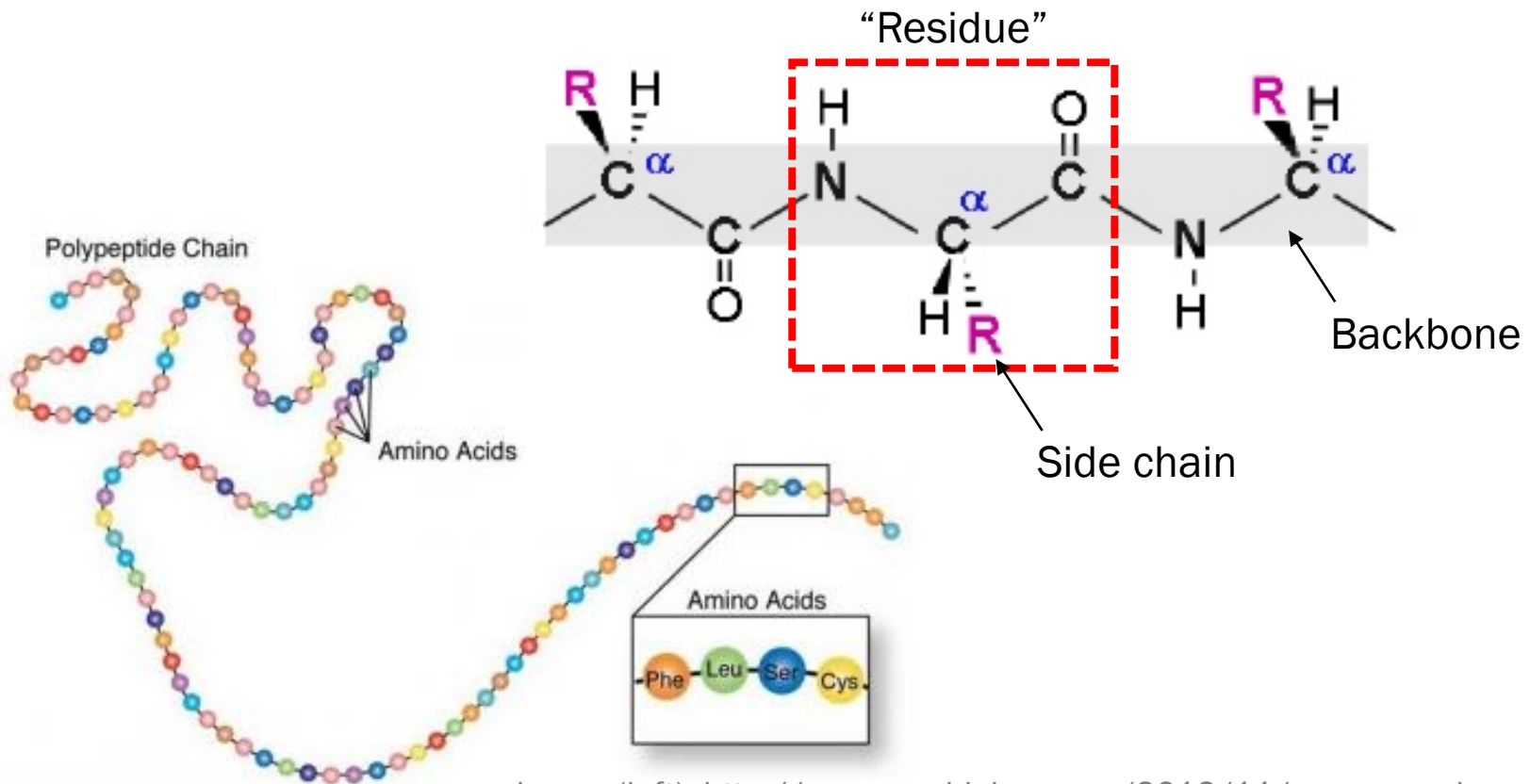


Image (left): <http://www.mcqbiology.com/2012/11/mcq-on-amino-acids.html>

Image (right): <https://www.slideshare.net/MUBOSScz/14-proteins>

20 Typical Monomeric Units

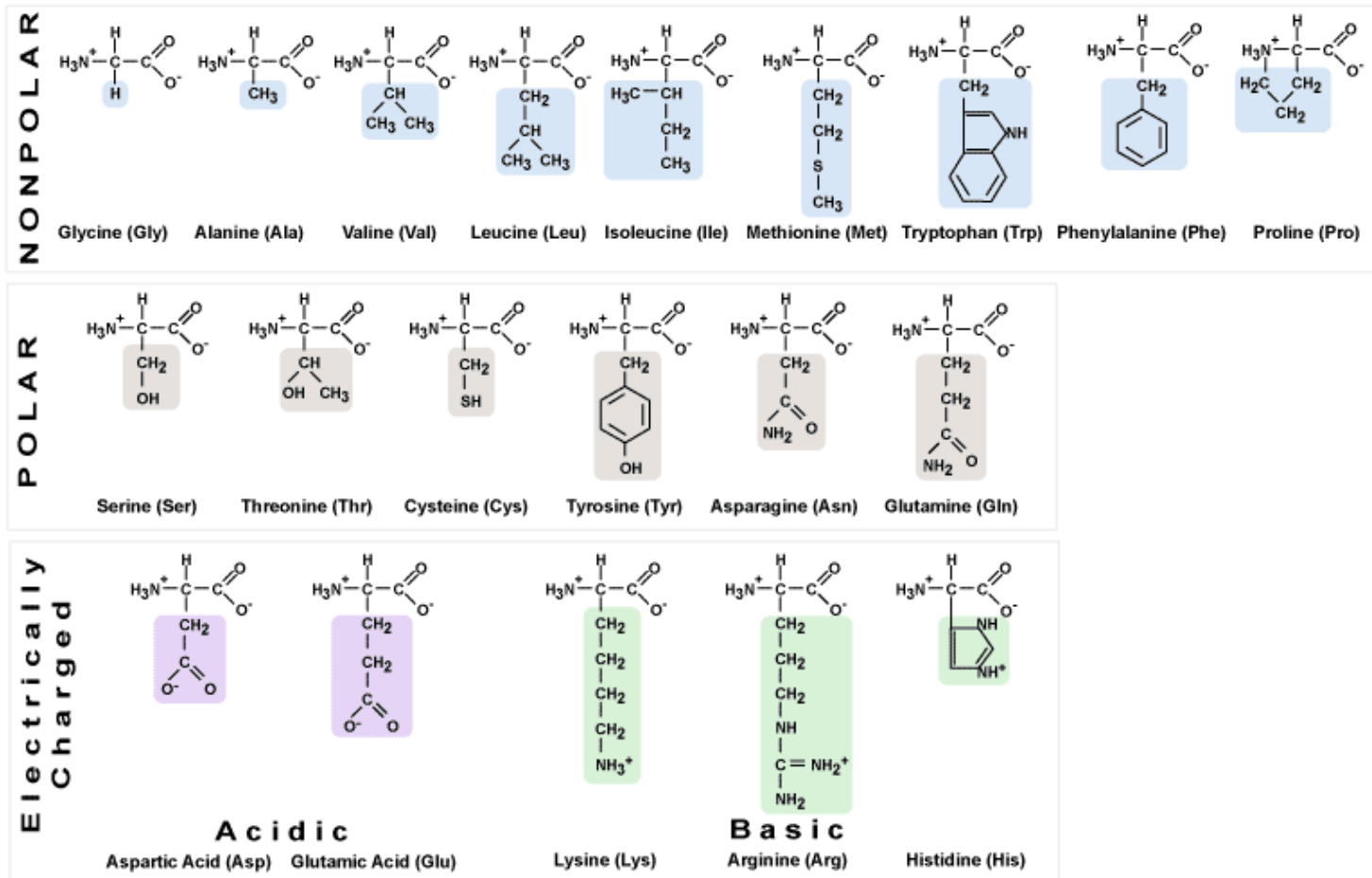


Image: <https://nuraisyahsukor.weebly.com/class-of-amino-acid-nitrogen-cycle-urea-cycle.html>

Proteins are Polymer Chains

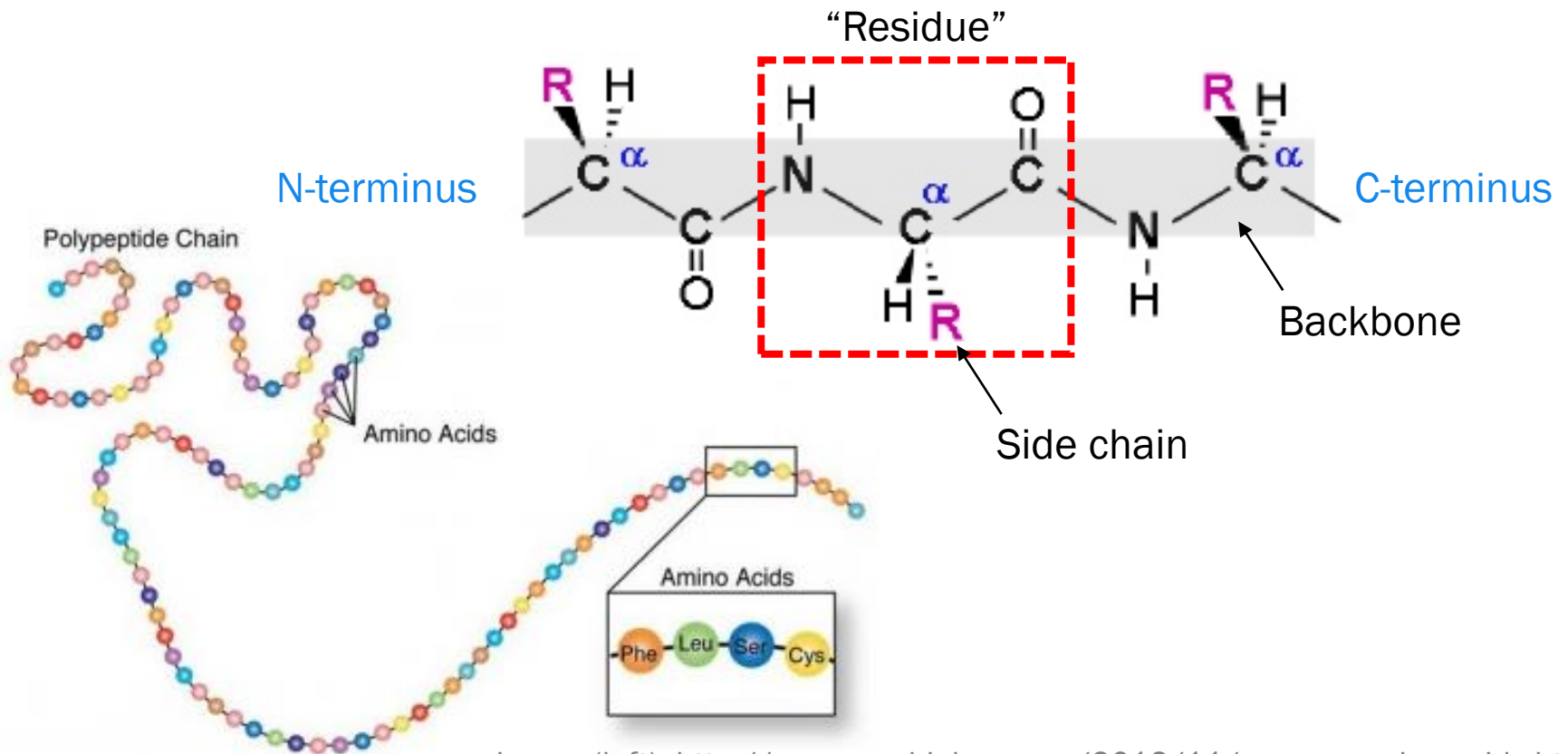
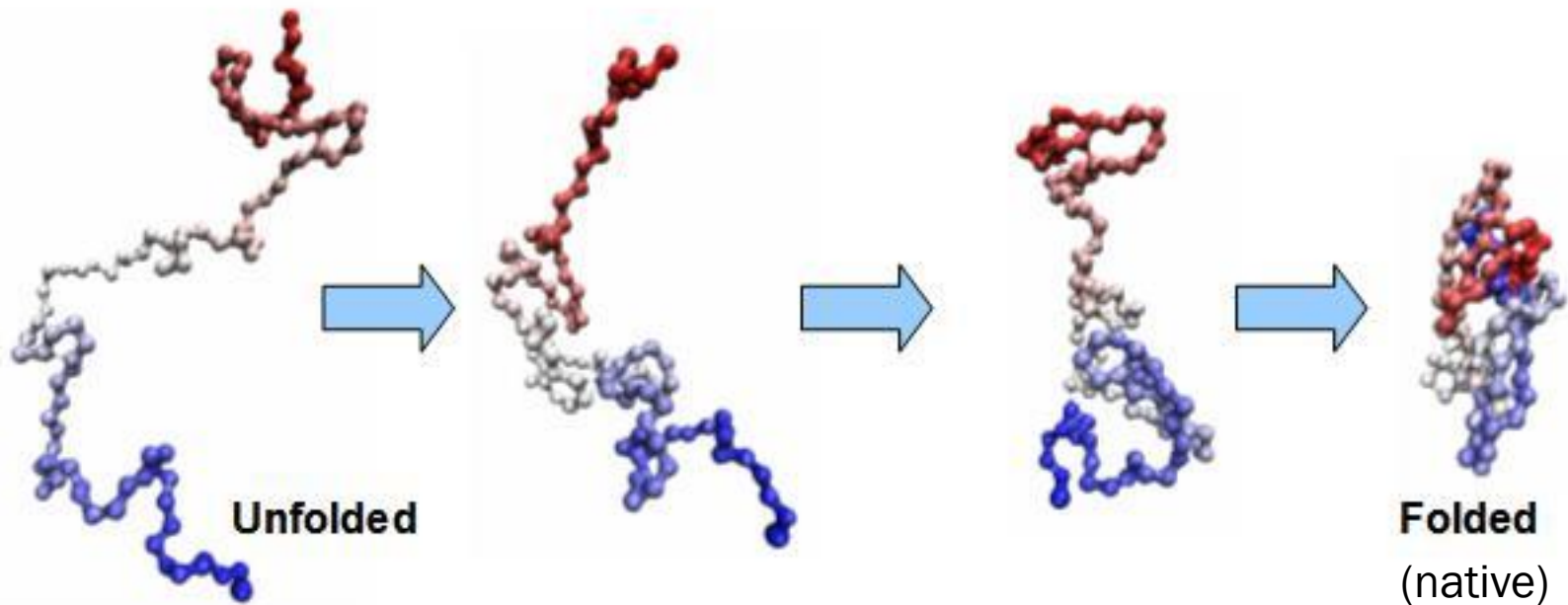


Image (left): <http://www.mcqbiology.com/2012/11/mcq-on-amino-acids.html>

Image (right): <https://www.slideshare.net/MUBOSScz/14-proteins>

Folding: How to Obtain 3D Structure

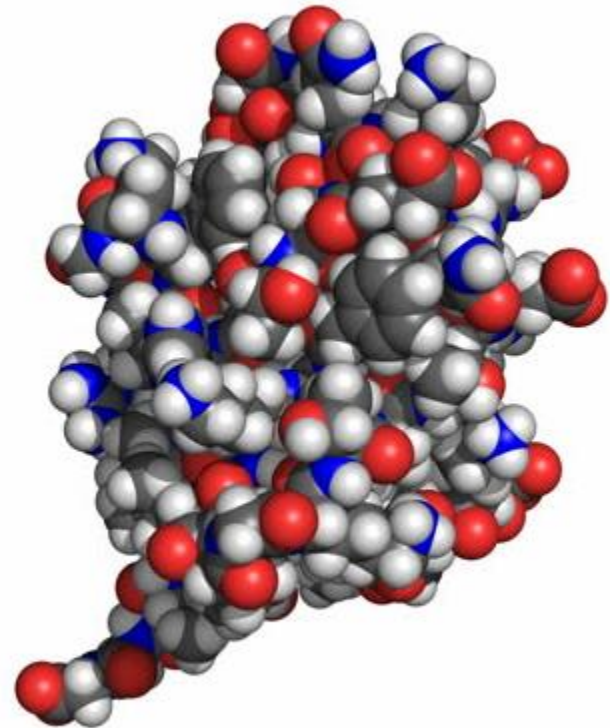


Seeing is believing: <https://youtu.be/yZ2aY5lxEGE>

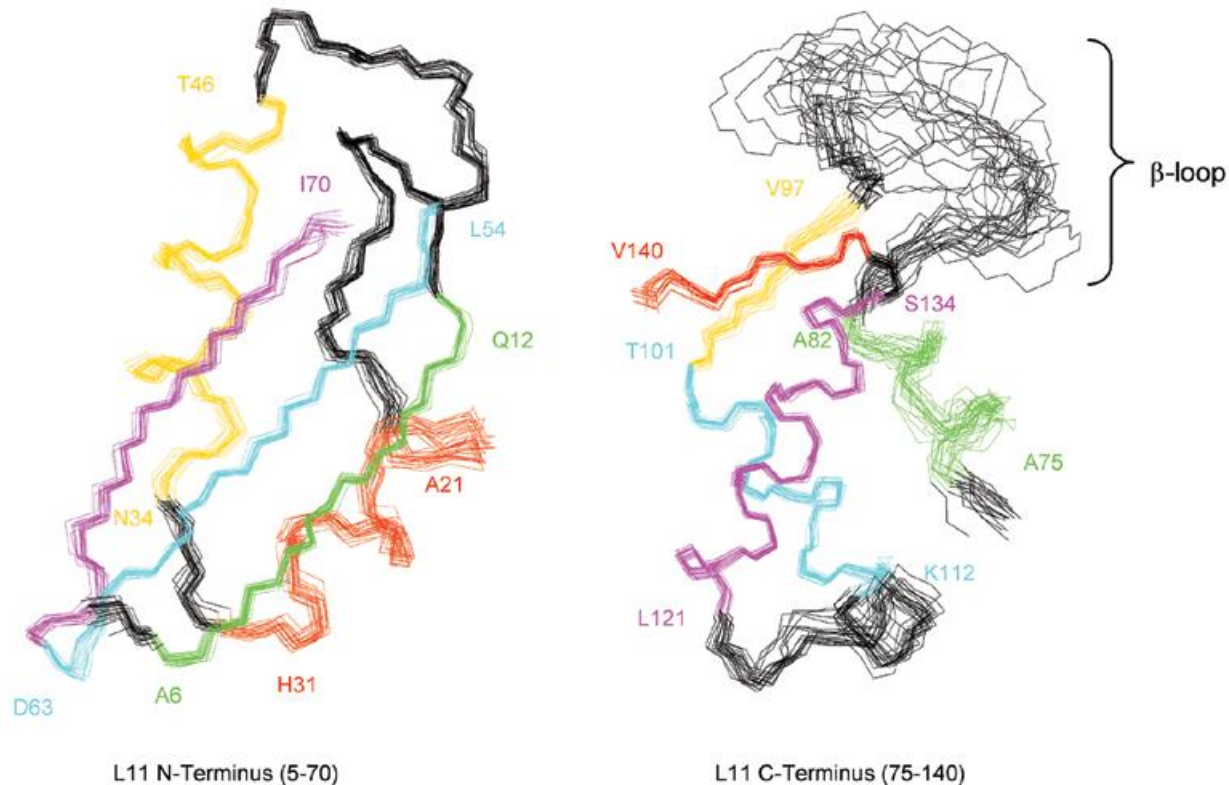
Image: https://inst.eecs.berkeley.edu/~ee127a/book/login/exa_protein_folding.html

Folded Structure is Dense

- Packing density
 - Organic solid: 0.68 – 0.8
 - Proteins: 0.72 – 0.77



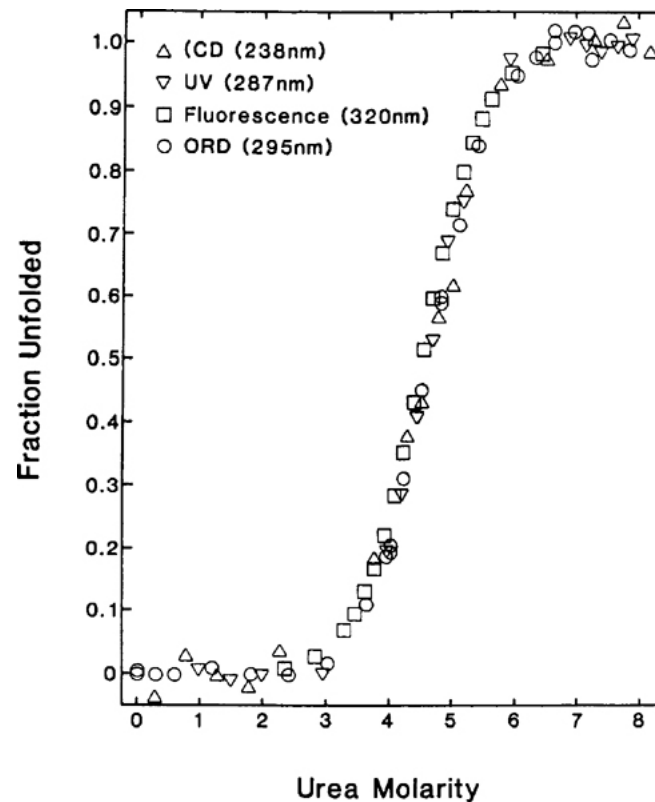
“Frozen” in the Folded State



(measurement by NMR)

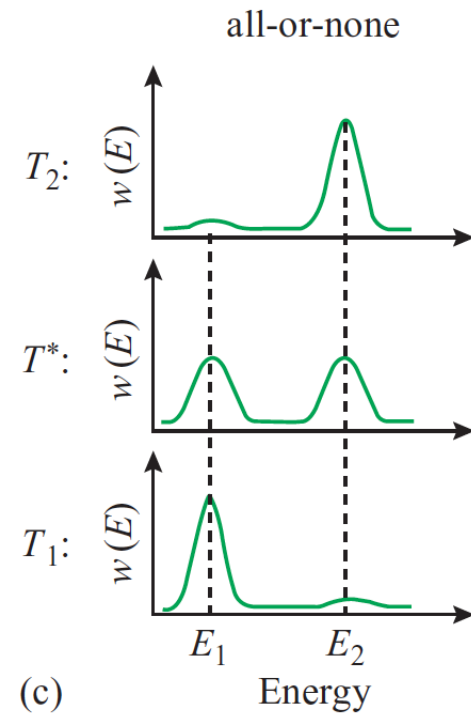
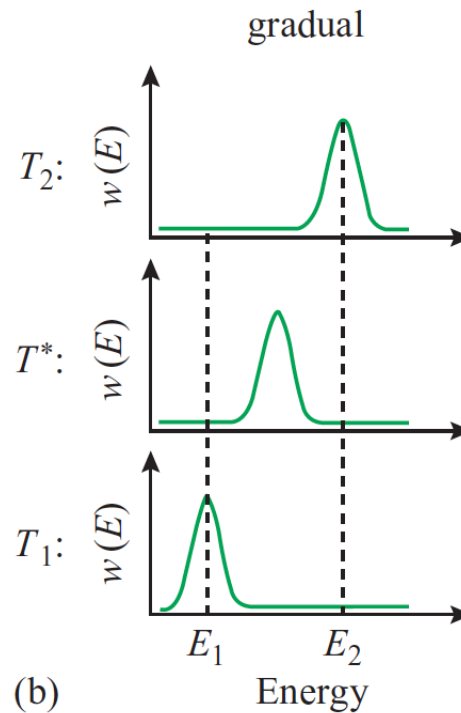
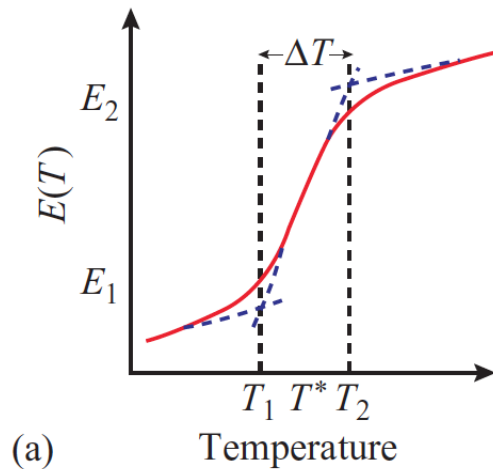
Image: <http://schwalbe.org.chemie.uni-frankfurt.de/node/684>

Protein Denaturation: S-shaped Curve



Shirley, "Urea and Guanidine Hydrochloride Denaturation Curves," in *Protein Stability and Folding* (1995)

Phase Transition?



van't Hoff Criterion

Assuming two-state folding,

$$P_{\text{fold}} = \frac{e^{-\beta G_{\text{folded}}}}{e^{-\beta G_{\text{folded}}} + e^{-\beta G_{\text{unfolded}}}} = \frac{1}{1 + e^{\beta \Delta G_{\text{fold}}}}$$

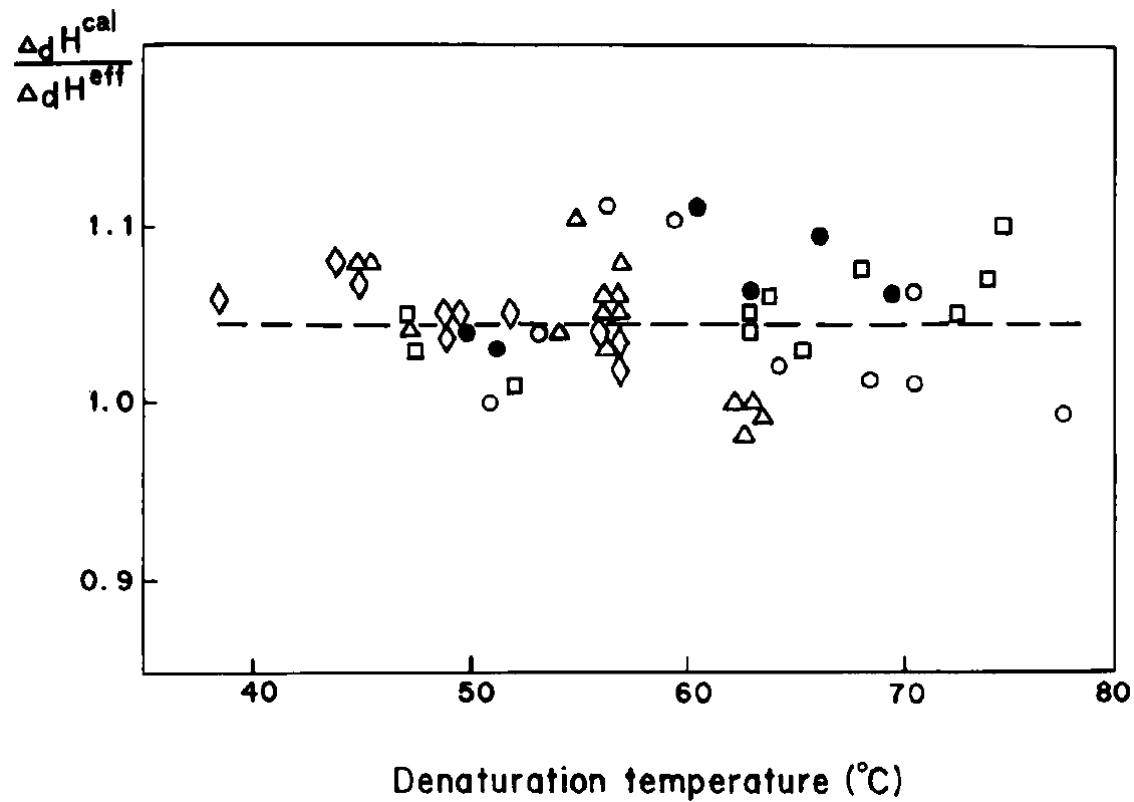
Using the Gibbs-Helmholtz equation,

$$\frac{dP_{\text{fold}}}{dT} = P_{\text{fold}}(1 - P_{\text{fold}}) \frac{\Delta H_{\text{fold}}}{k_B T^2}$$

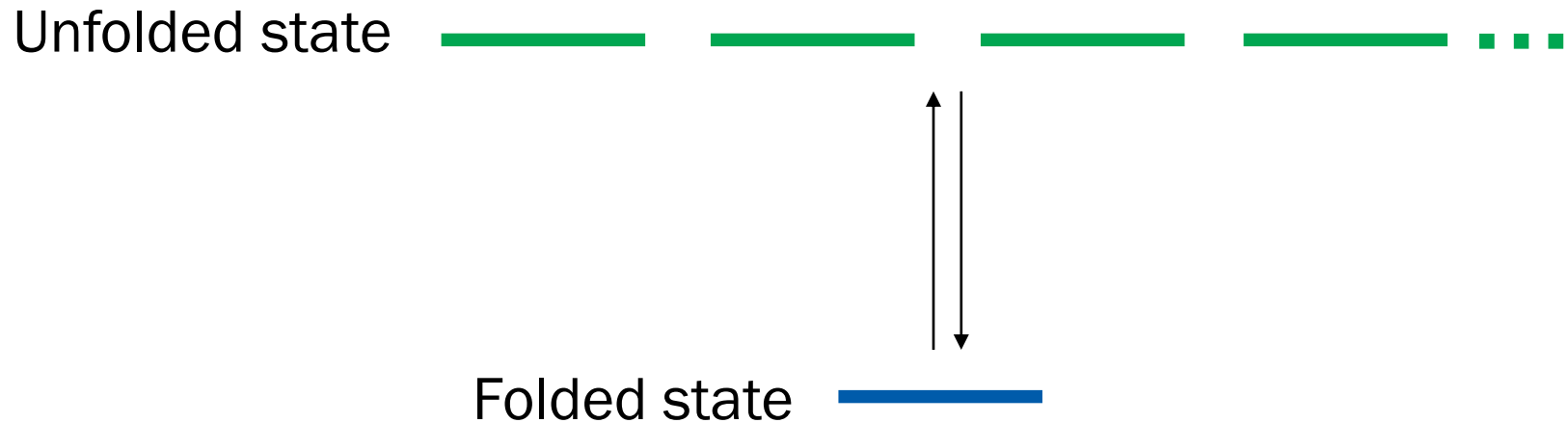
At the transition point ($P_{\text{fold}} = 1/2$),

$$\left. \frac{dP_{\text{fold}}}{dT} \right|_{T=T_f} = \frac{\Delta H_{\text{fold}}}{4k_B T_f^2}$$

Experimentally ...



Two-State Model



What is the underlying physics?

“Funnel” Picture

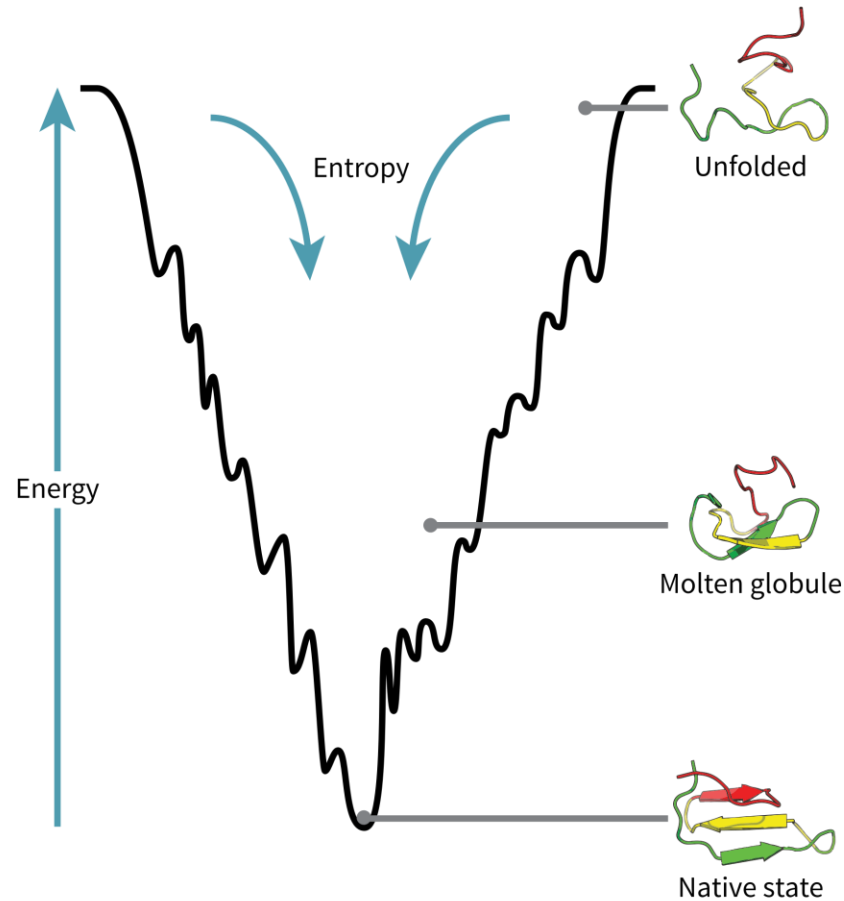
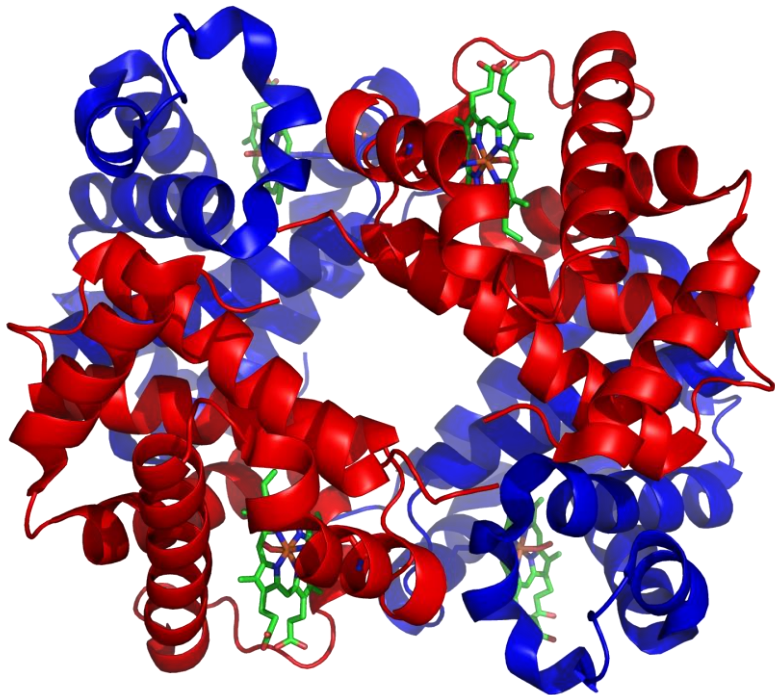


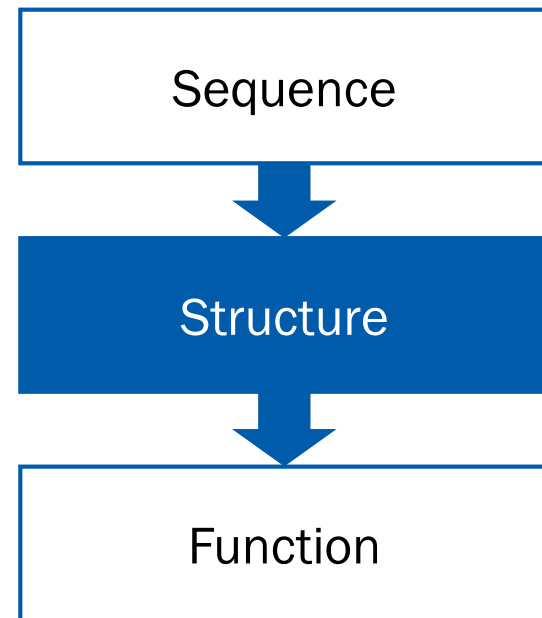
Image: https://en.wikipedia.org/wiki/Protein_folding#/media/File:Folding_funnel_schematic.svg

Why is Folding Important?

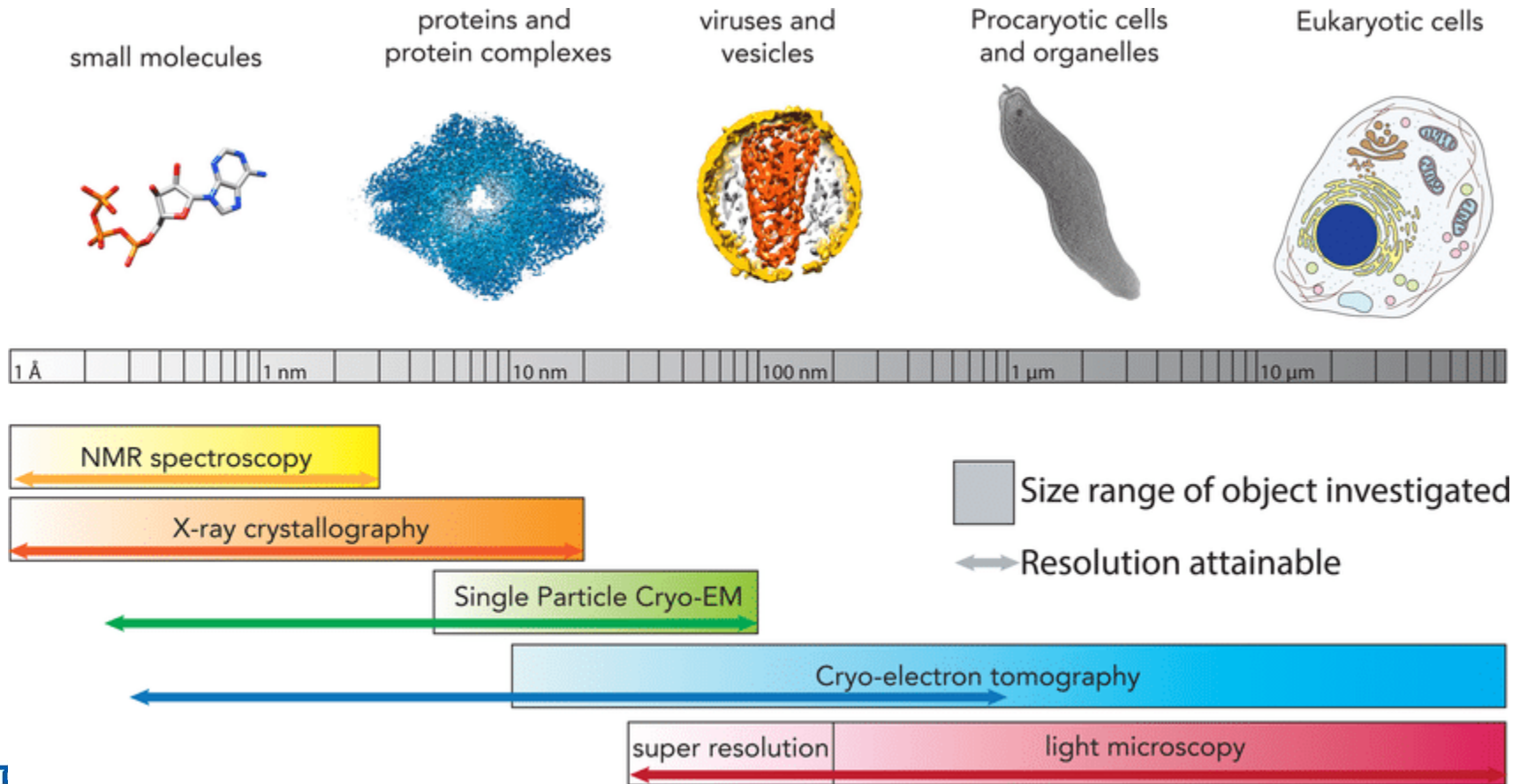


Hemoglobin

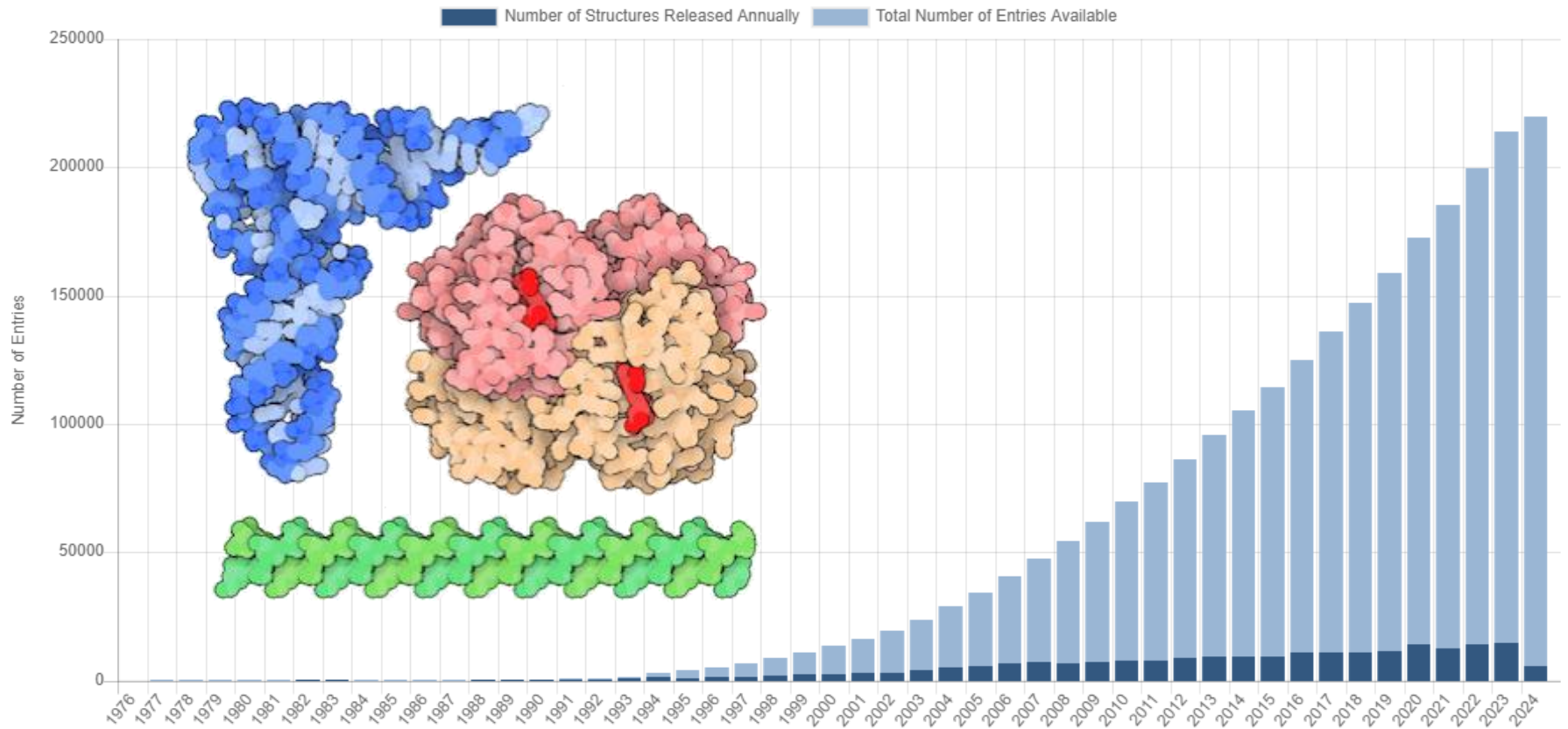
Structure-function paradigm



Structural Biology

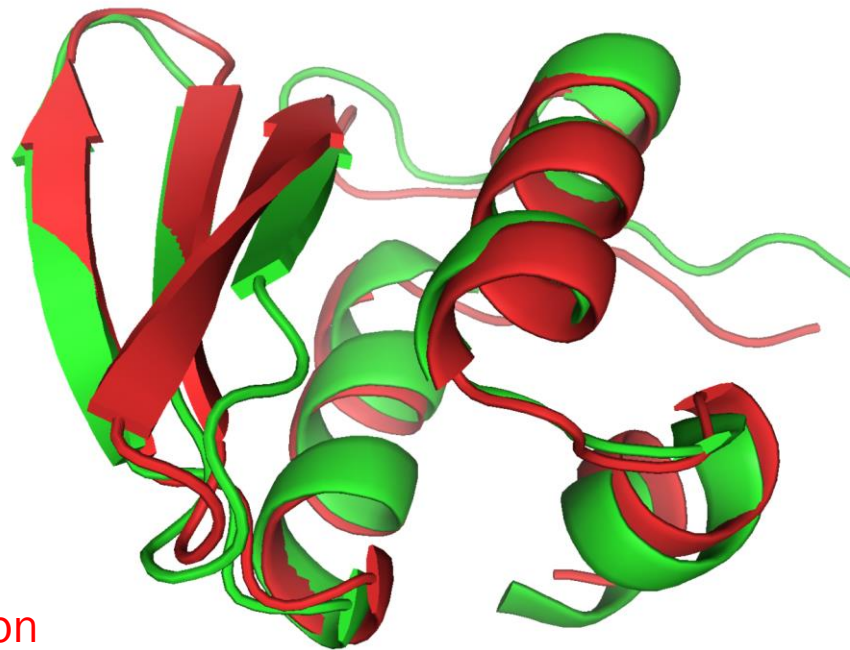


PDB: Structural Database



<https://www.rcsb.org/stats/growth/growth-released-structures>

“Computational” Structural Biology

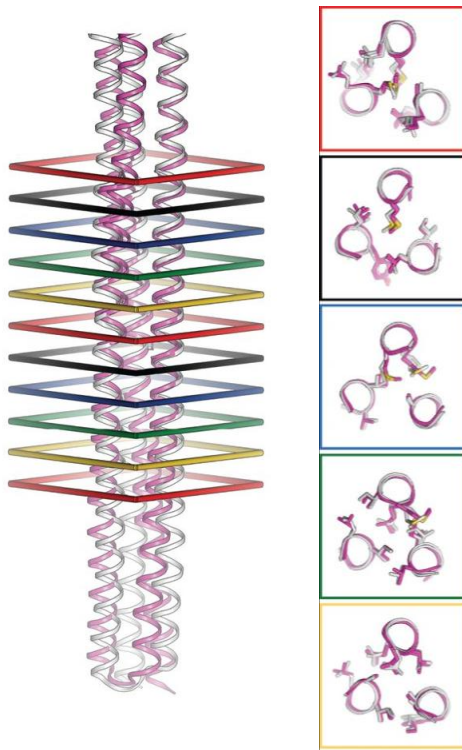


Red: prediction

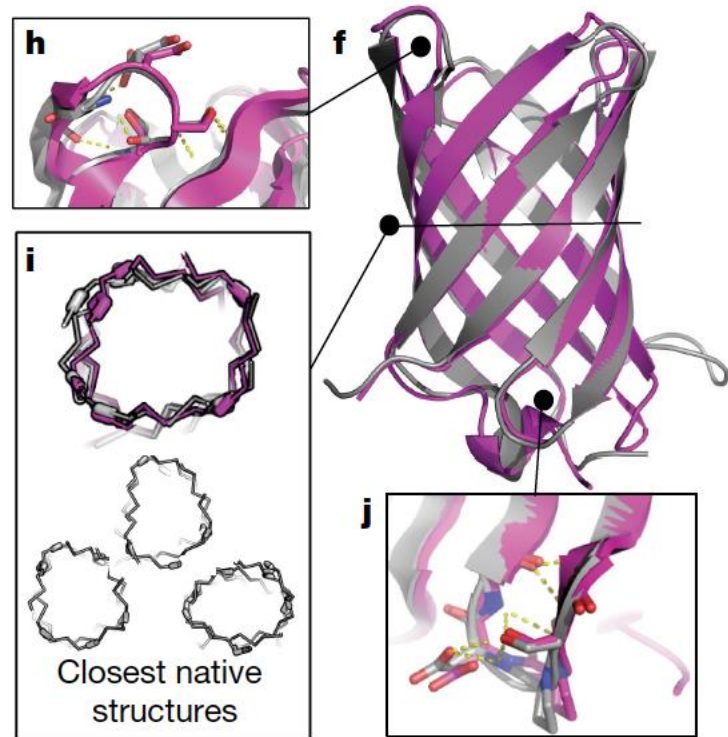
Green: experiment

De novo Structure Design

α structure



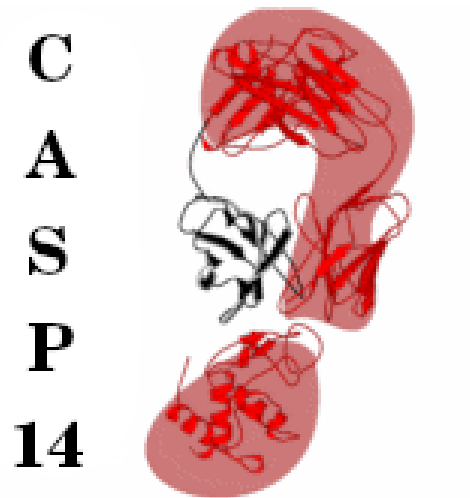
β structure



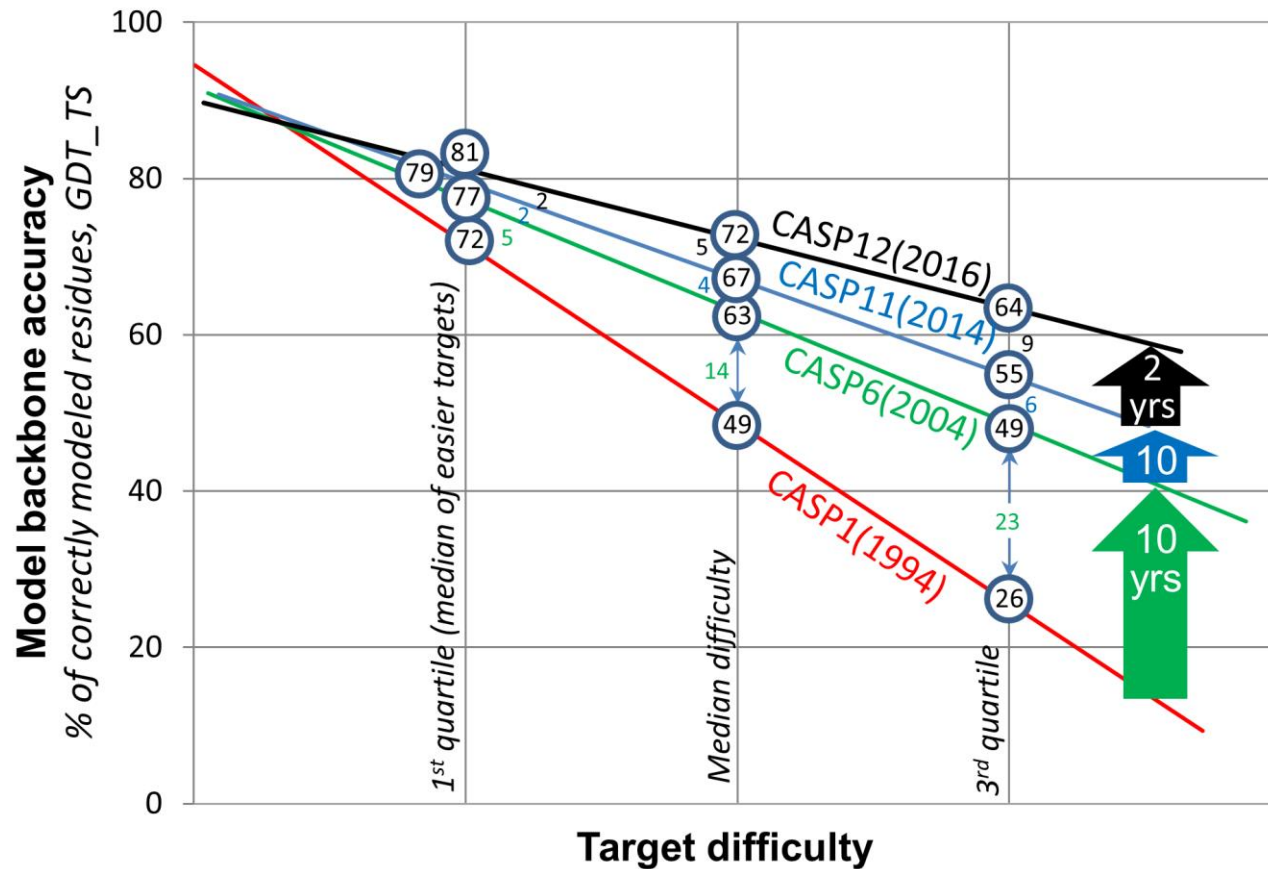
Huang et al., *Science* (2014); Dou et al., *Nature* (2018)

CASP

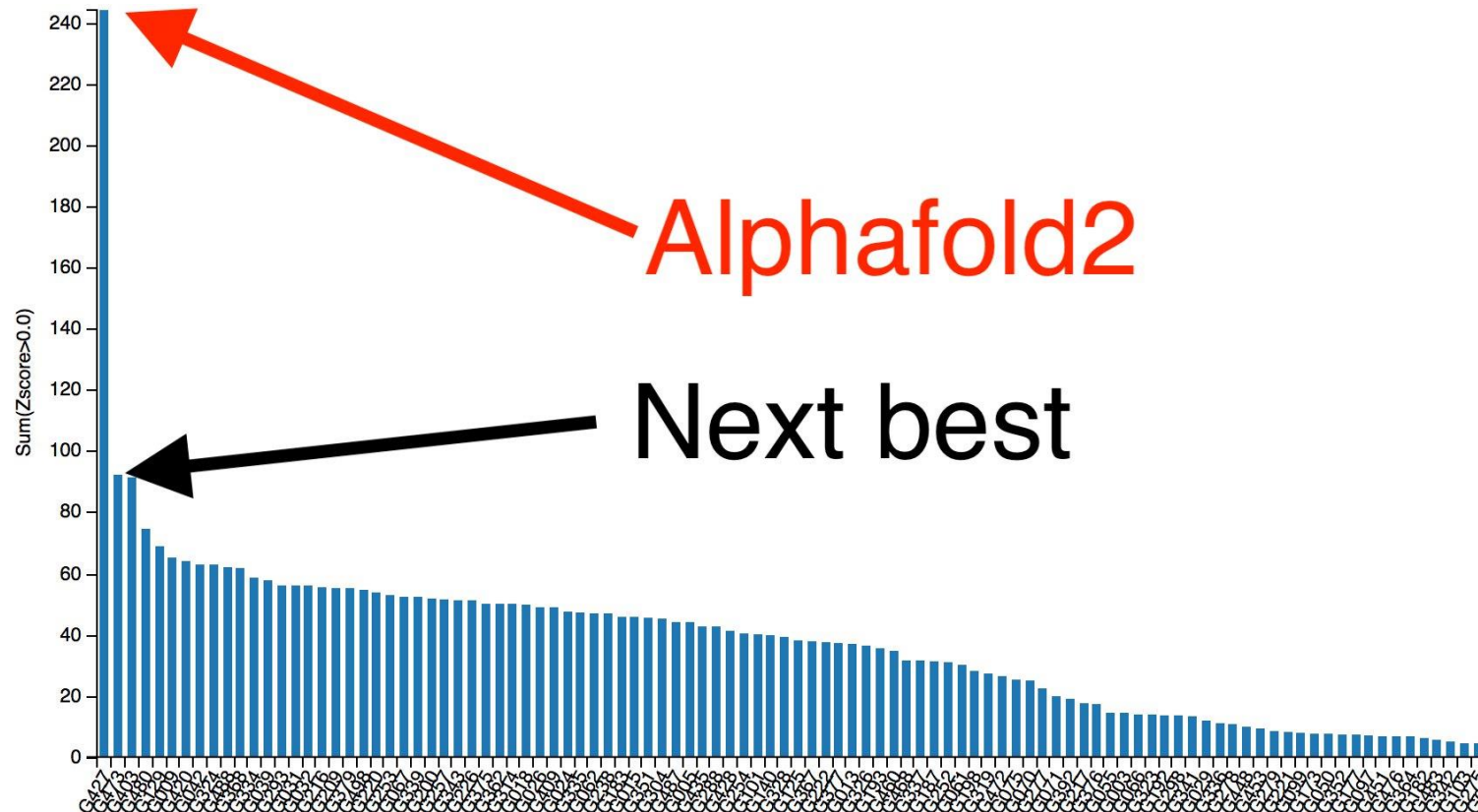
- Critical Assessment of protein Structure Prediction
(<http://predictioncenter.org/>)



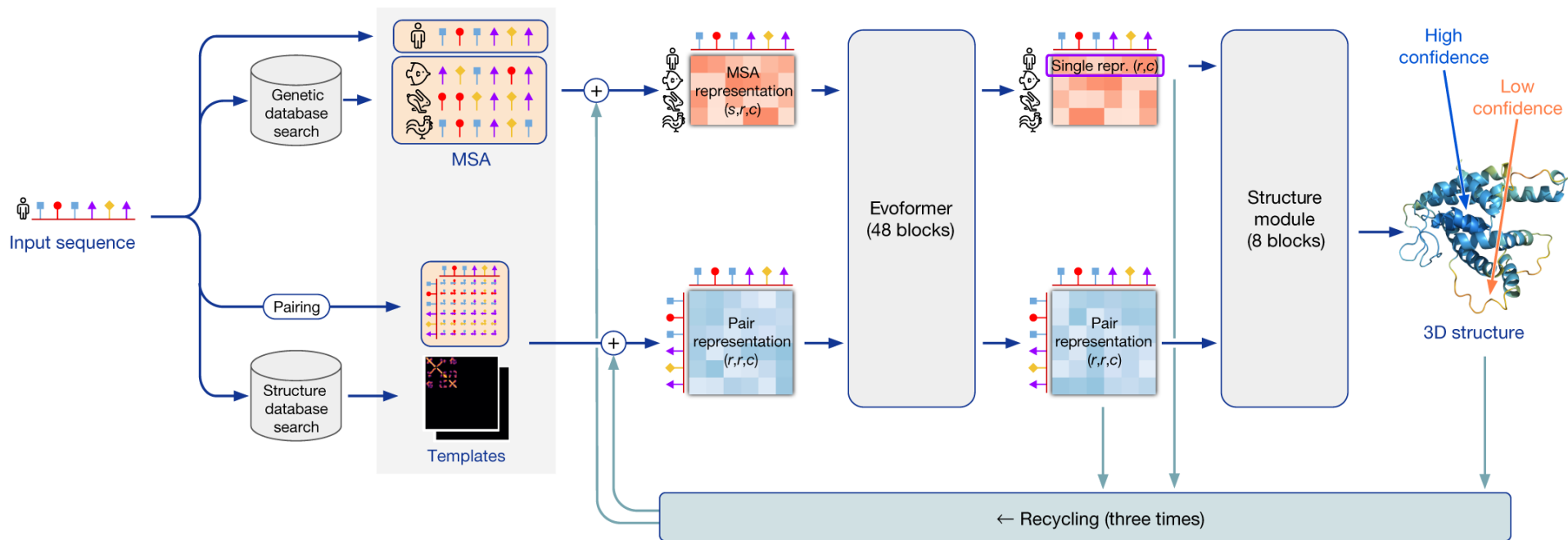
Accuracy of Prediction Models



Shocking News from CASP14 (2020)



AlphaFold 2



Thermodynamics of Folding

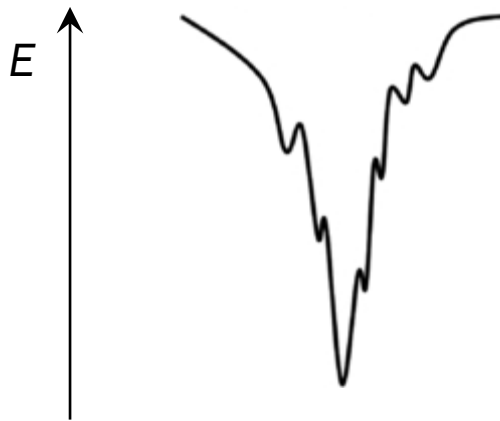


Image (top): <https://www.quantamagazine.org/how-disordered-proteins-are-upending-molecular-biology-20170118>

Image (bottom): Mészáros et al., *Phys Biol* (2011)

What If ...

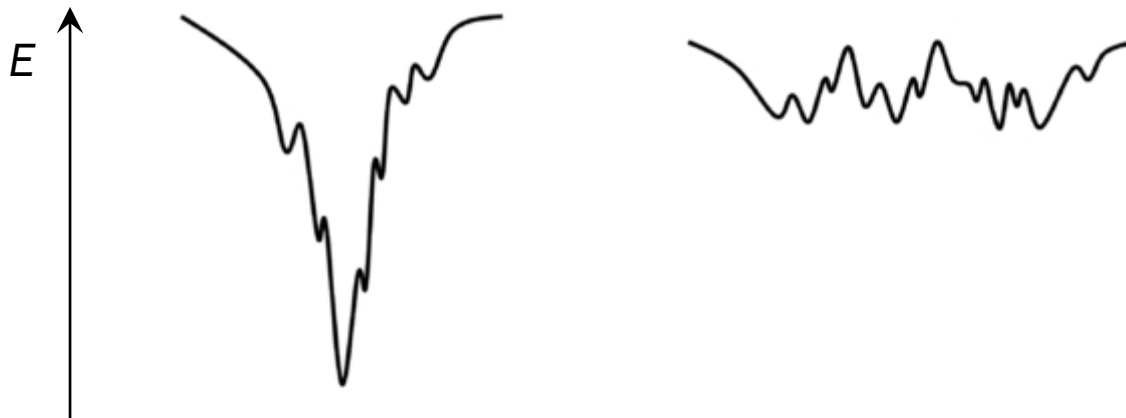


Image (top): <https://www.quantamagazine.org/how-disordered-proteins-are-upending-molecular-biology-20170118>

Image (bottom): Mészáros et al., *Phys Biol* (2011)

No Fixed Structure

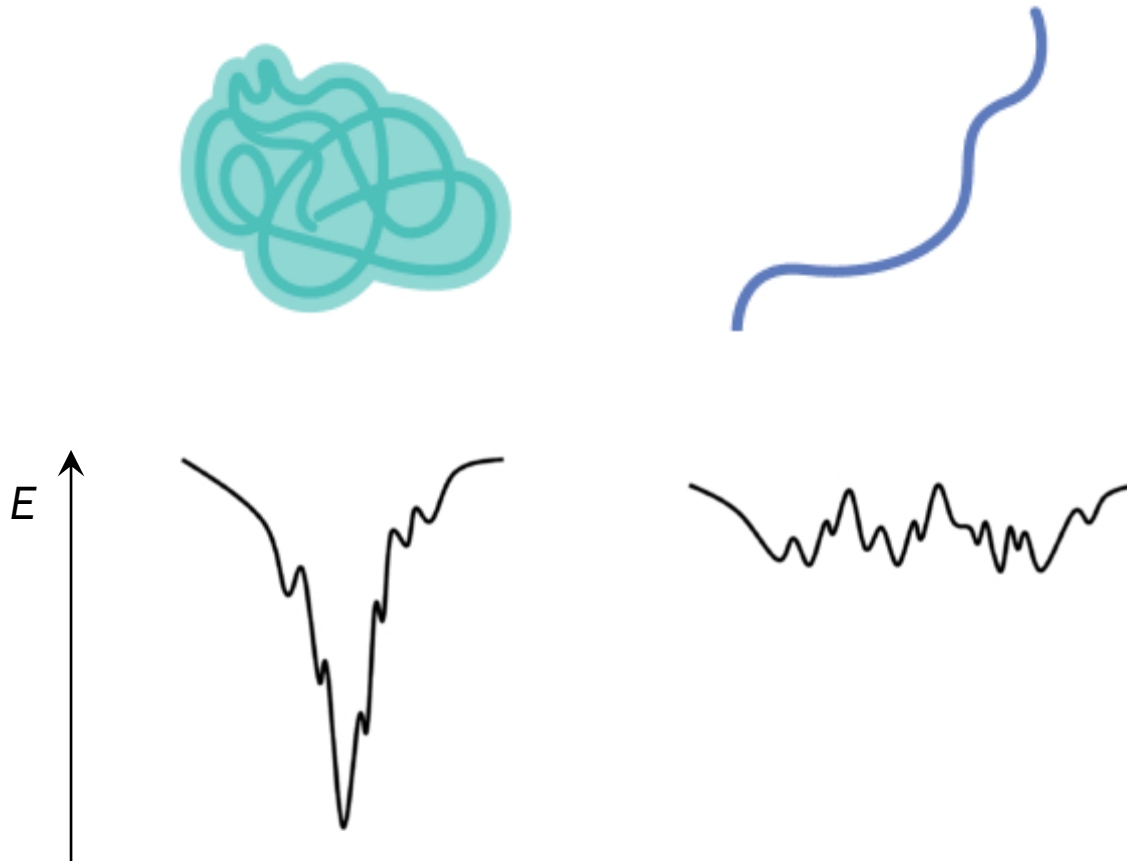


Image (top): <https://www.quantamagazine.org/how-disordered-proteins-are-upending-molecular-biology-20170118>

Image (bottom): Mészáros et al., *Phys Biol* (2011)

Intrinsically Disordered Protein (IDP)



foldable protein



intrinsically disordered protein

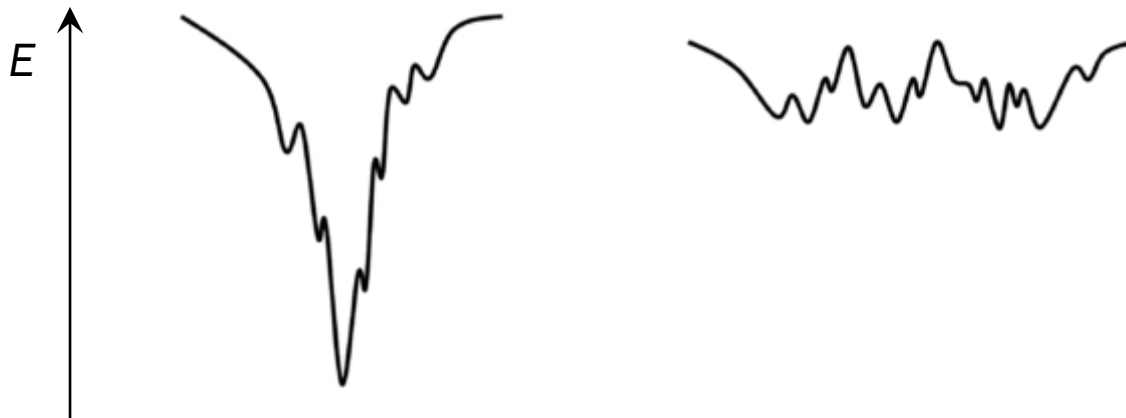
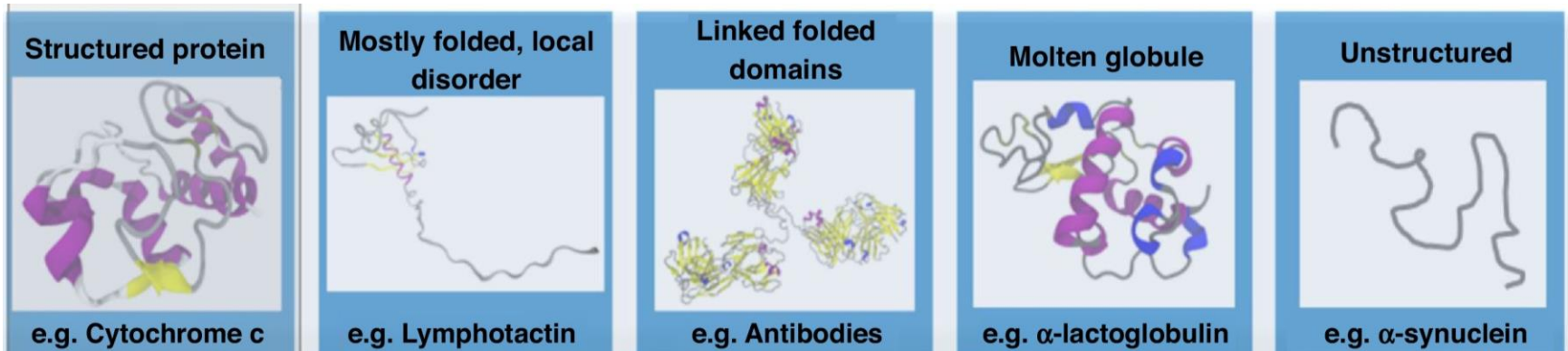


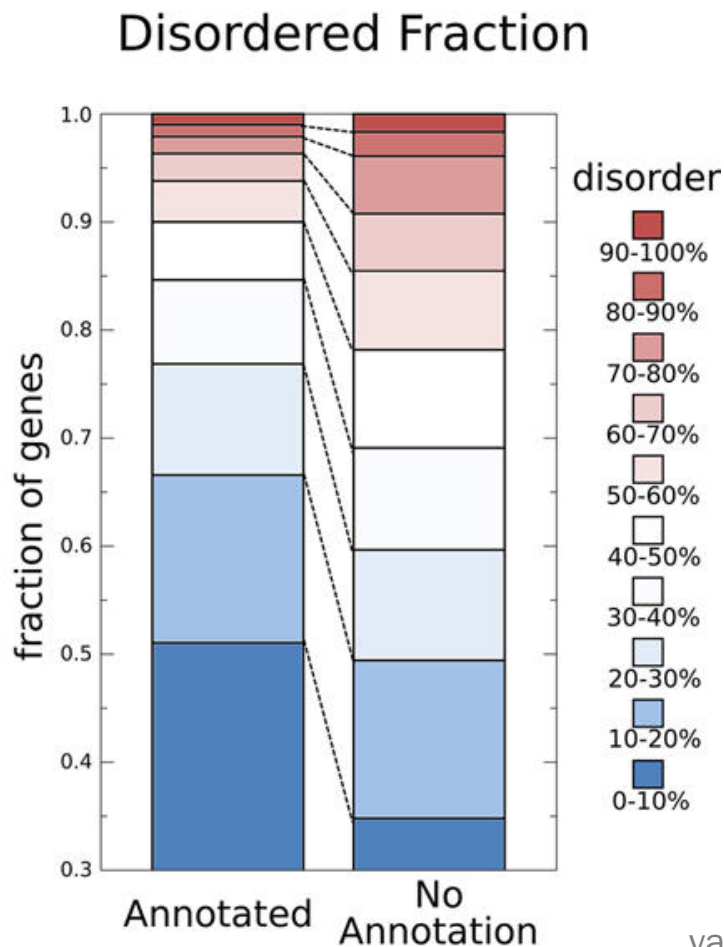
Image (top): <https://www.quantamagazine.org/how-disordered-proteins-are-upending-molecular-biology-20170118>

Image (bottom): Mészáros et al., *Phys Biol* (2011)

Spectrum of IDPs

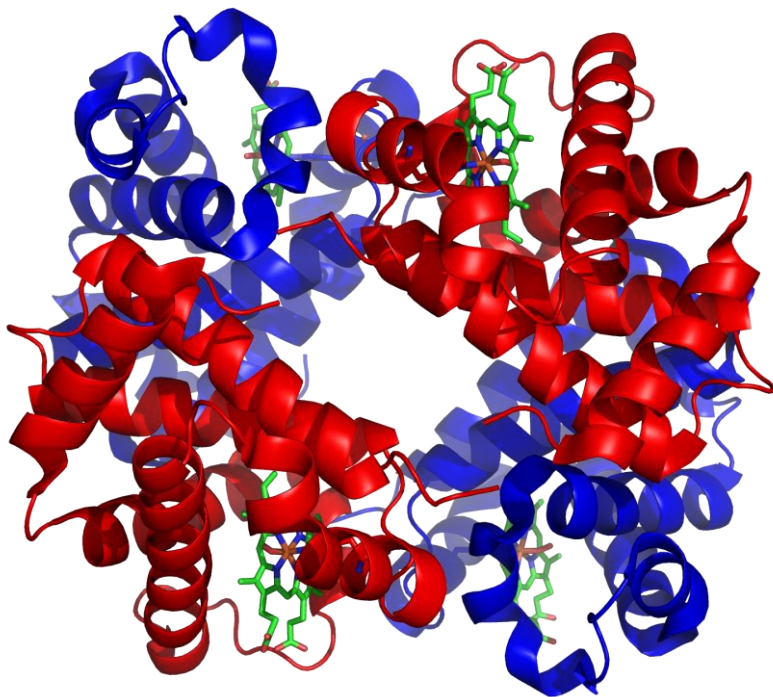


IDPs are Prevalent *in vivo*



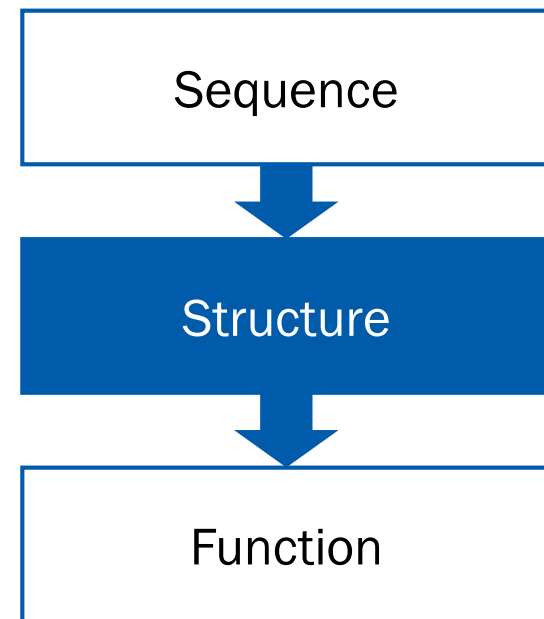
van der Lee *et al.*, *Chem Rev* (2014)

Functions of “Foldable” Proteins



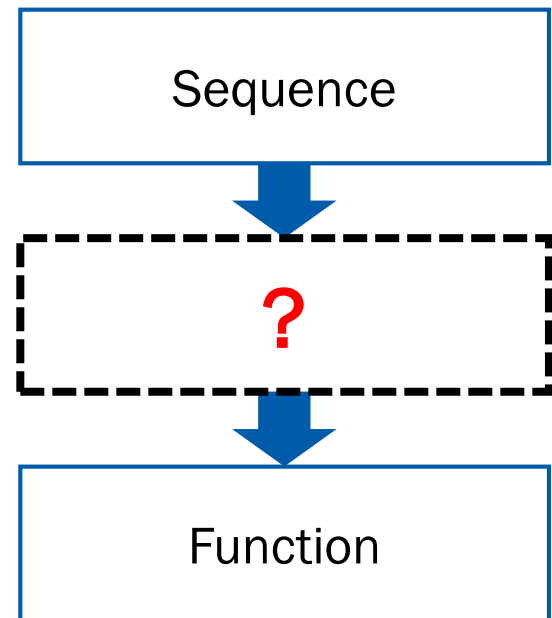
Hemoglobin

Structure-function paradigm



How do IDPs Obtain Functions?

Structure-function paradigm (?)



Working Principles of IDPs

Entropic chains

Coupled folding
and binding

Fibril formation

Phase separation

Working Principles of IDPs

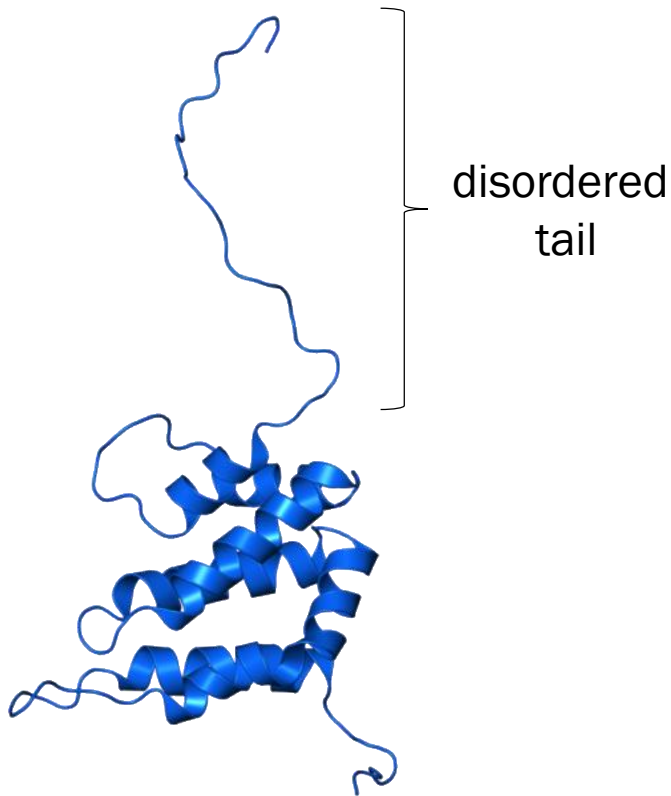
Entropic chains

Coupled folding
and binding

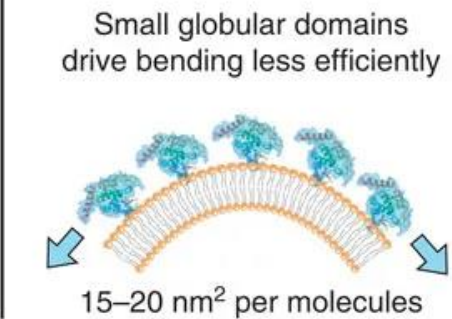
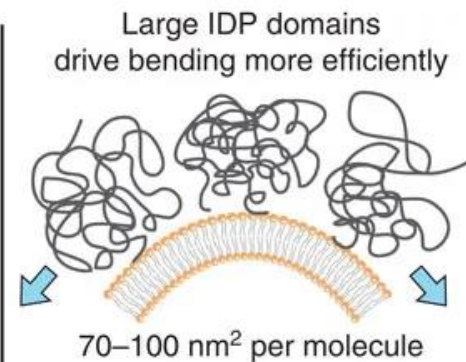
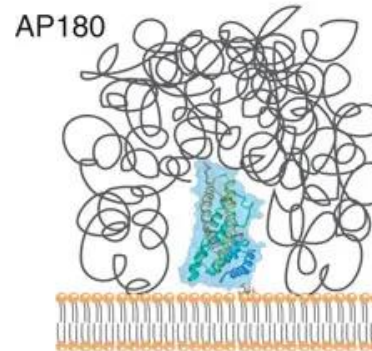
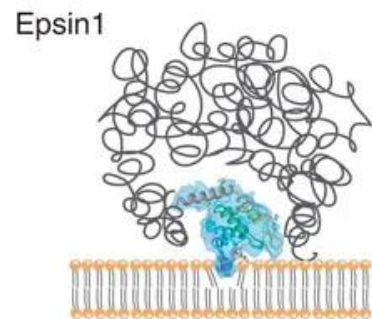
Fibril formation

Phase separation

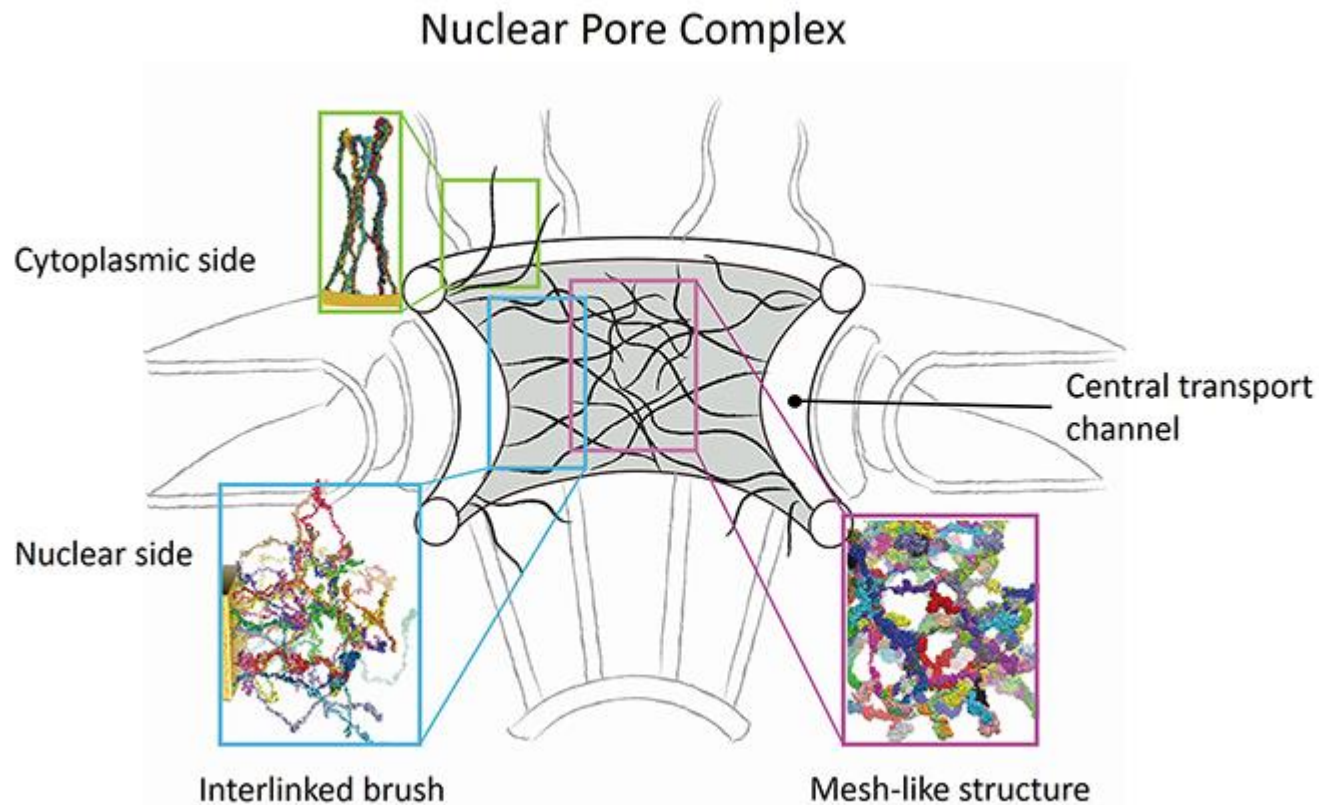
Entropic Chains



Epsin1



Entropic Chains: FG Repeats



Working Principles of IDPs

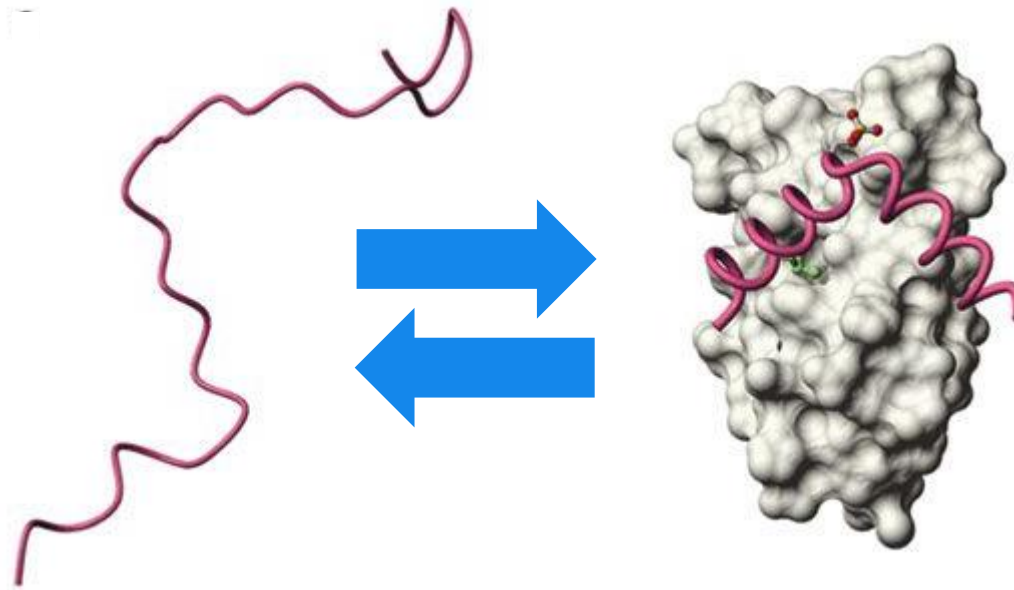
Entropic chains

Coupled folding
and binding

Fibril formation

Phase separation

Coupled Folding and Binding



pKID-KIX interaction

Working Principles of IDPs

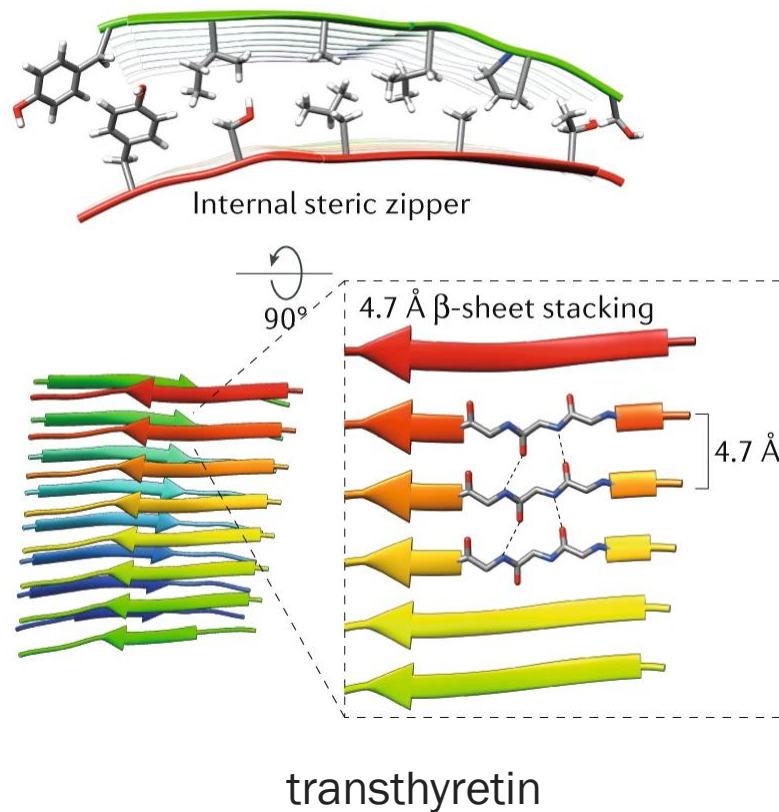
Entropic chains

Coupled folding
and binding

Fibril formation

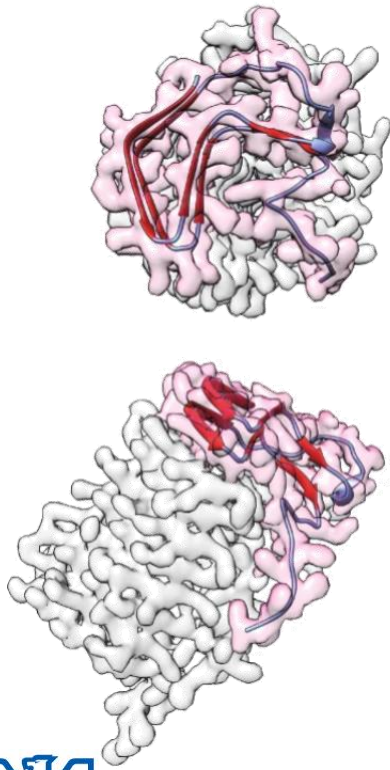
Phase separation

Fibril Formation

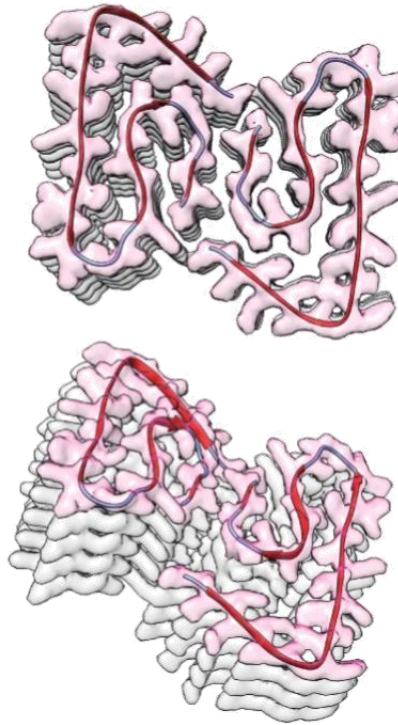


Diversity in Fibril Structures

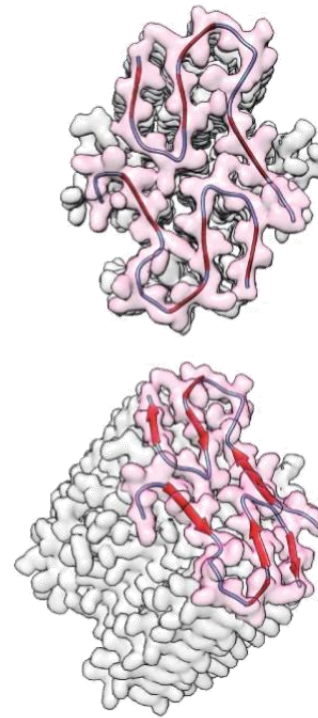
a HET-S
(2lbu)



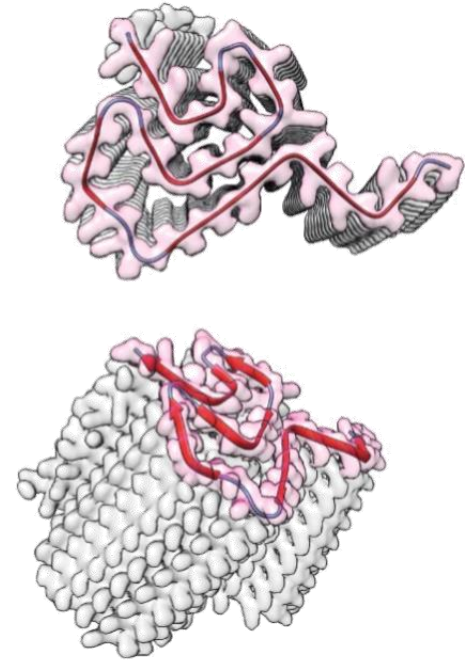
b A β 1-42 'LS'
(5oqv)



A β 1-42
(5kk3)

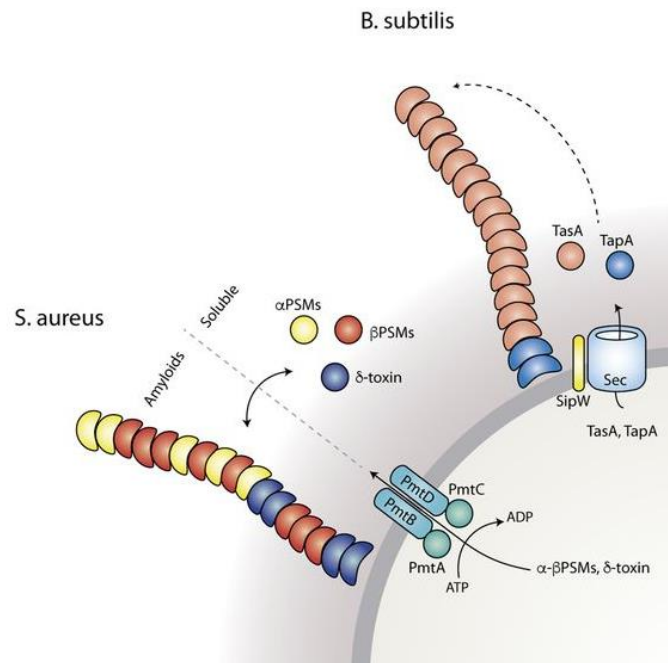


e α -Synuclein
(2n0a)

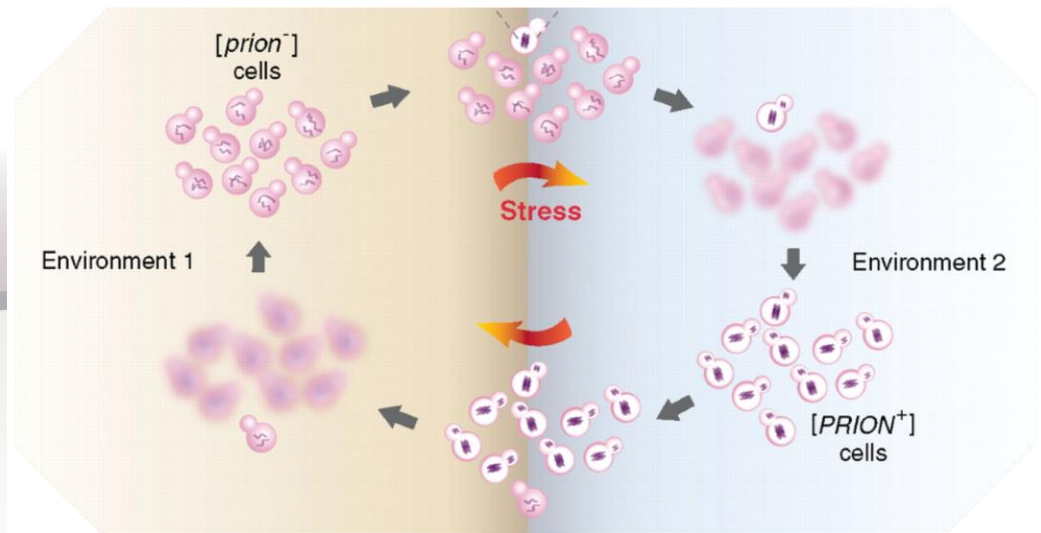


Biology: Functional Amyloids

Biofilm formation



Epigenetic switch



Biology: Neurodegenerative Diseases

Alzheimer's disease

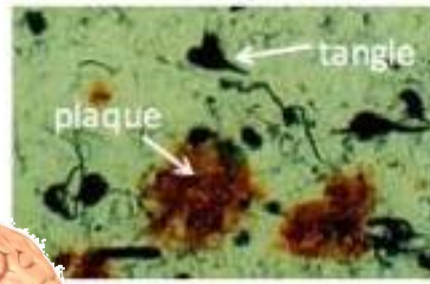
amyloid beta, tau

Parkinson's disease

α -synuclein

Huntington's disease

huntingtin



BSE (mad cow disease)

prion

ALS (Lou Gehrig's disease)

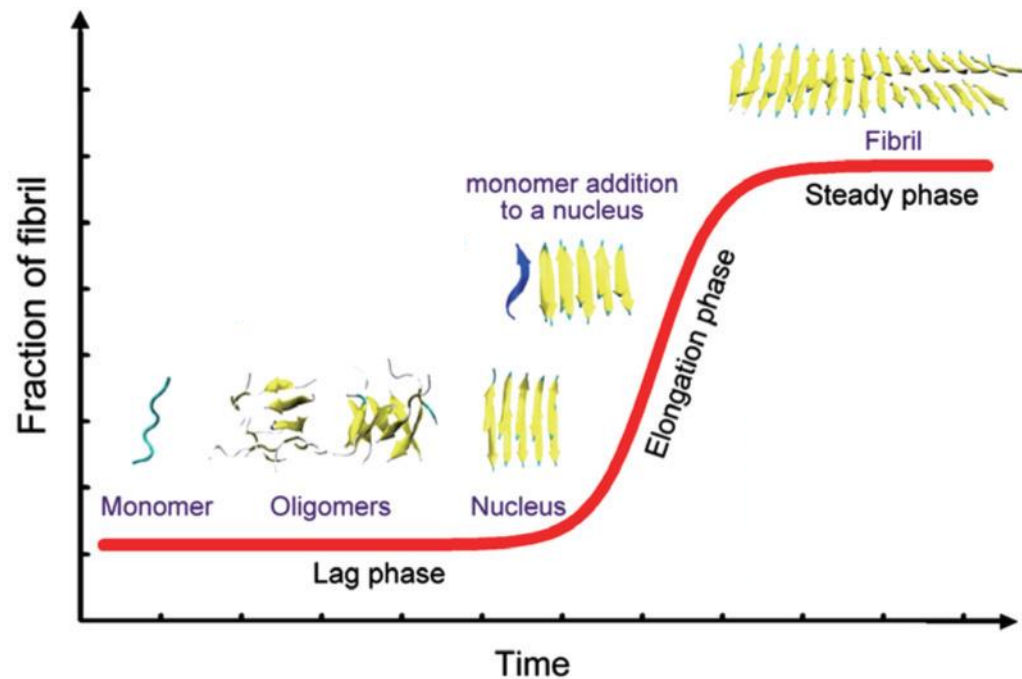
TDP43, FUS



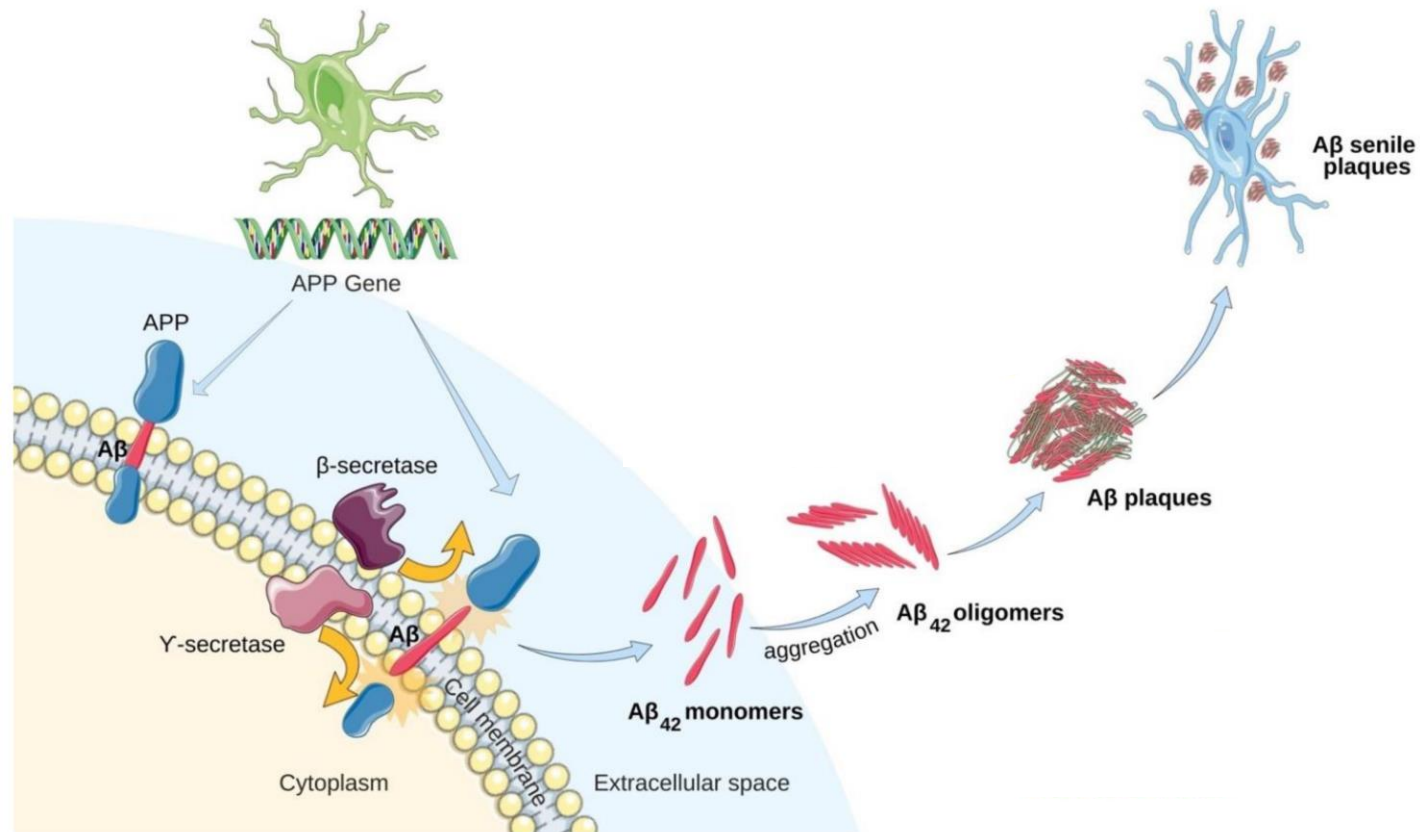
Images: <https://www.livescience.com/29365-human-brain.html>

http://www.chemistry.emory.edu/faculty/lynn/Yan_Liang.html

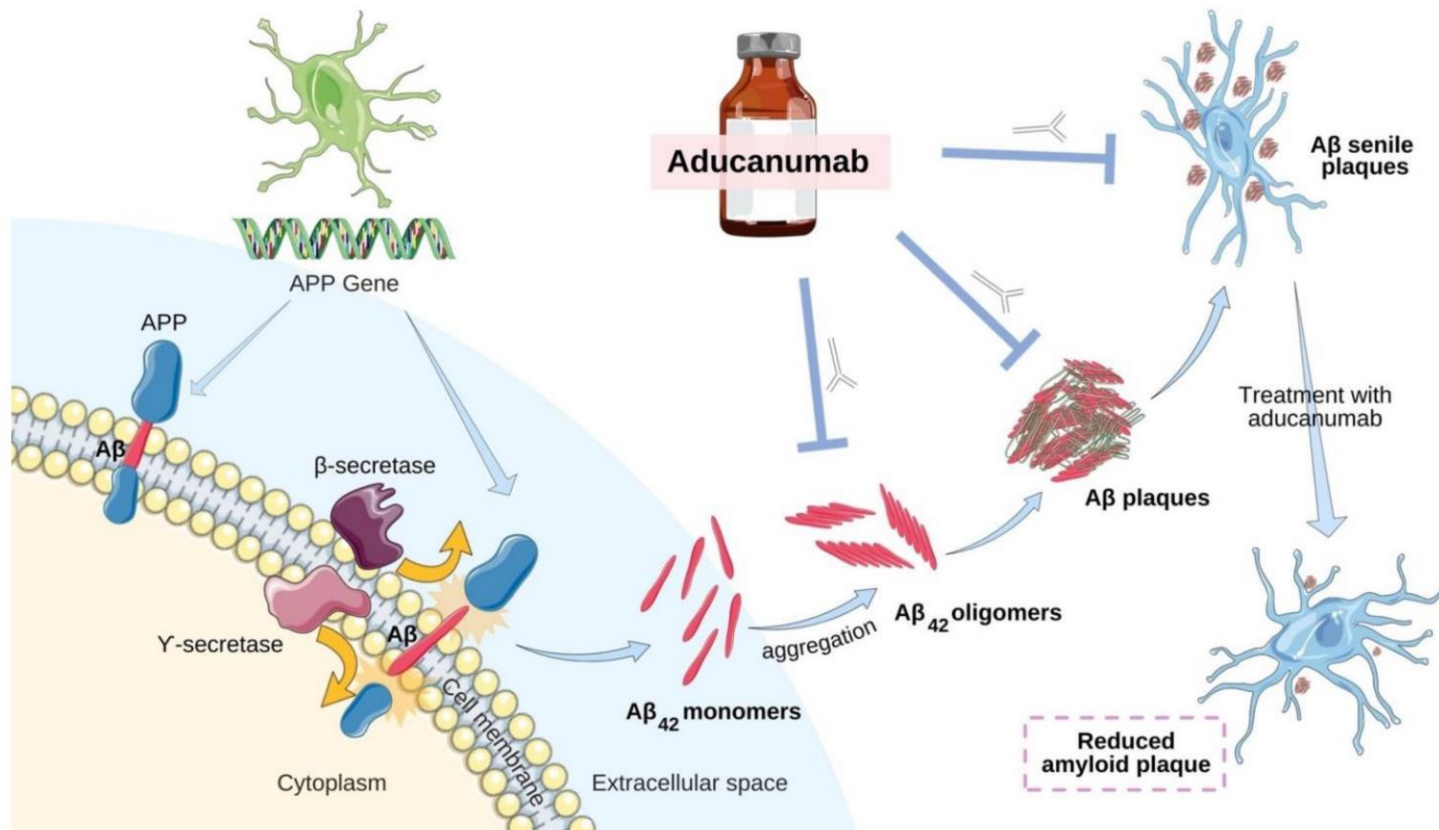
Mechanism of Fibrillation



Amyloid Cascade Hypothesis



Aducanumab (Aduhelm)



Working Principles of IDPs

Entropic chains

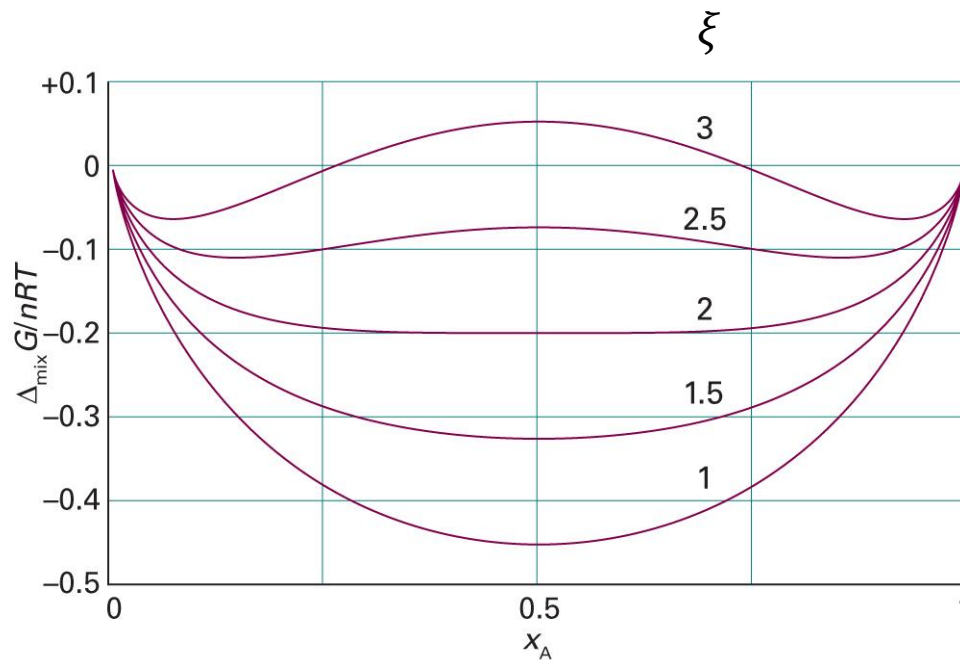
Coupled folding
and binding

Fibril formation

Phase separation

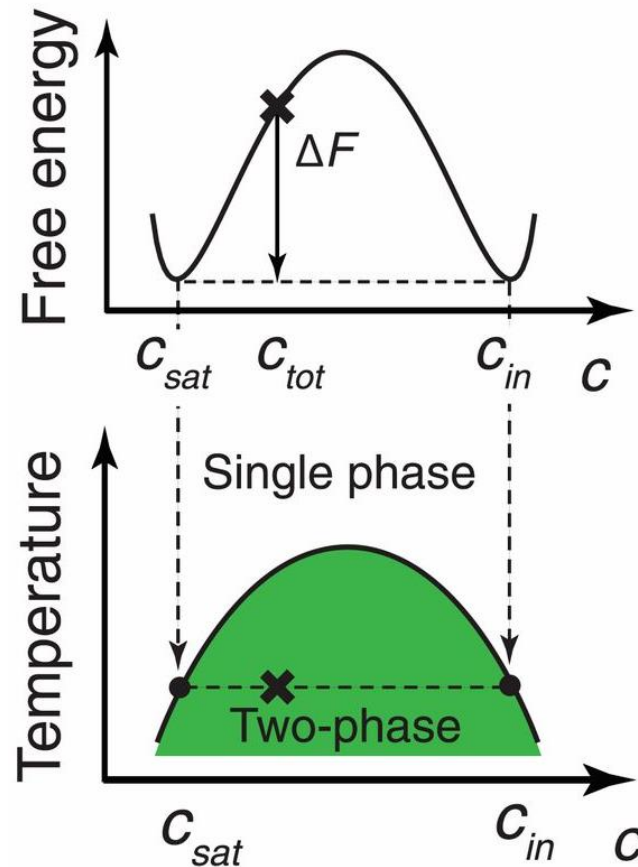
BW Model: Gibbs Energy of Mixing

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B + \xi x_A x_B)$$



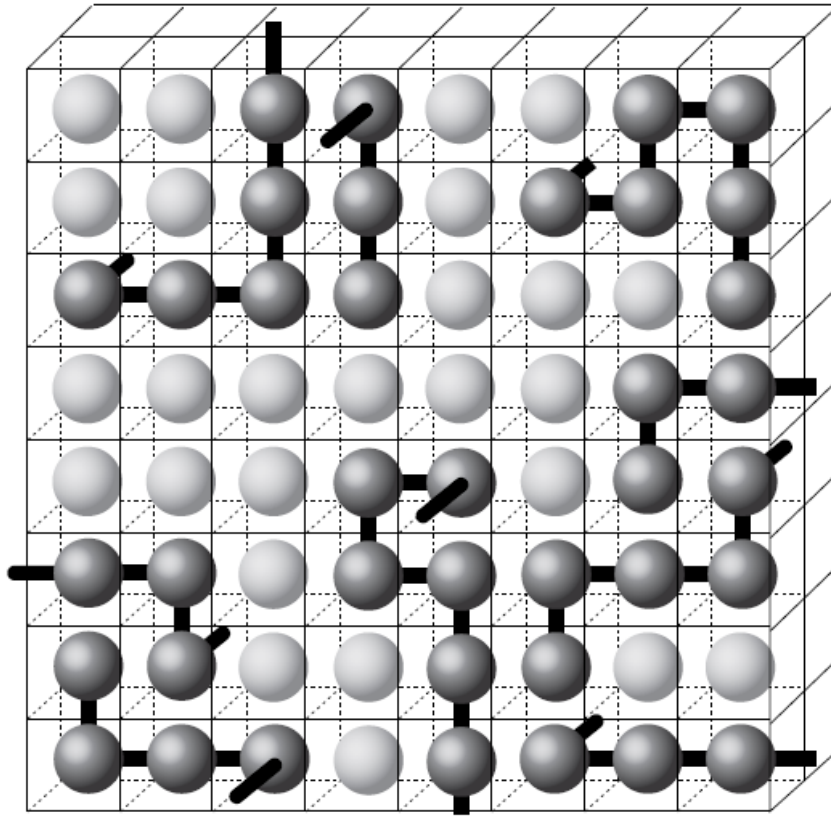
(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5B.5)

What Happens at $\xi > 2$?



(Image: Shin and Brangwynne, *Science* 357: eaaf4382 (2017), Figure 3)

Lattice Model of Polymers



M sites

N_s solvent molecules

N_p polymer molecules

L monomers / polymer

Entropy of Mixing

$$\Delta_{\text{mix}}S = -k_B \left(N_s \ln \frac{N_s}{M} + N_p \ln \frac{LN_p}{M} \right)$$

Define volume fractions:

$$\phi_s = \frac{N_s}{M}, \quad \phi_p = \frac{LN_p}{M}$$

Hence,

$$\frac{\Delta_{\text{mix}}S}{M} = -k_B \left(\phi_s \ln \phi_s + \frac{\phi_p}{L} \ln \phi_p \right)$$

Thermodynamics of Mixing

- Entropy of mixing

$$\frac{\Delta_{\text{mix}}S}{M} = -k_B \left(\phi_s \ln \phi_s + \frac{\phi_p}{L} \ln \phi_p \right)$$

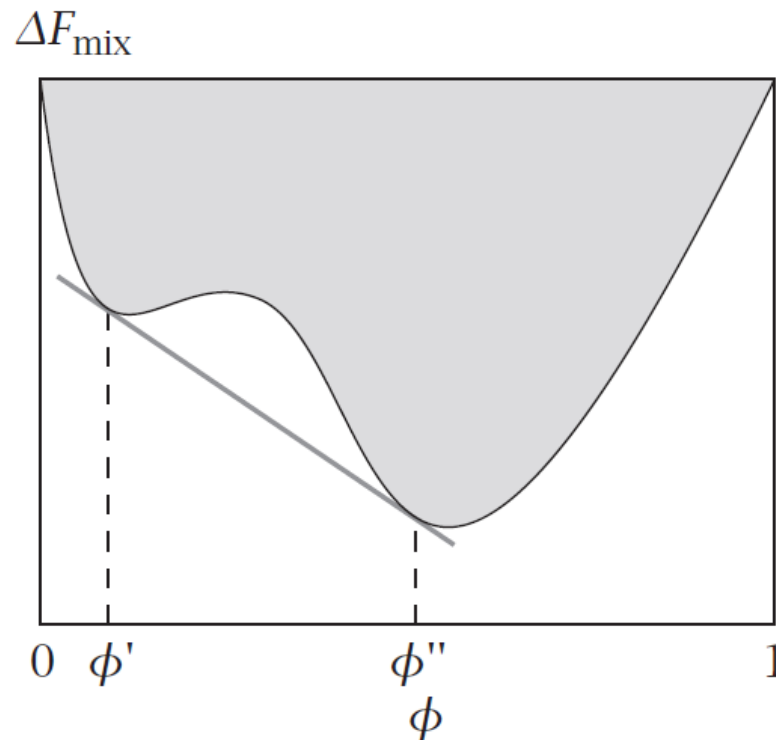
- Energy of mixing

$$\Delta_{\text{mix}}U = k_B T \xi_{\text{sp}} \frac{N_s N_p L}{M} = M k_B T \xi_{\text{sp}} \phi_s \phi_p$$

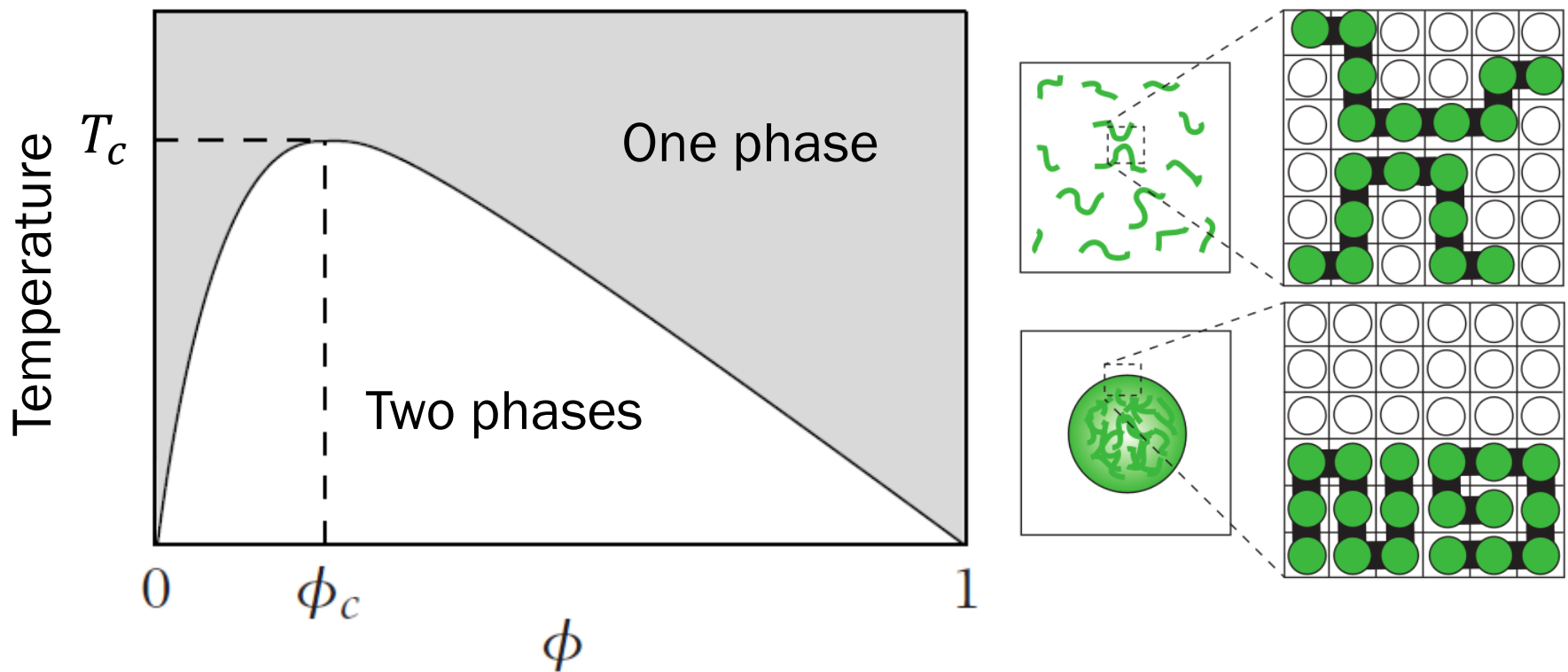
- Helmholtz energy of mixing

$$\frac{\Delta_{\text{mix}}F}{M k_B T} = \phi_s \ln \phi_s + \frac{\phi_p}{L} \ln \phi_p + \xi_{\text{sp}} \phi_s \phi_p$$

Free Energy of Mixing

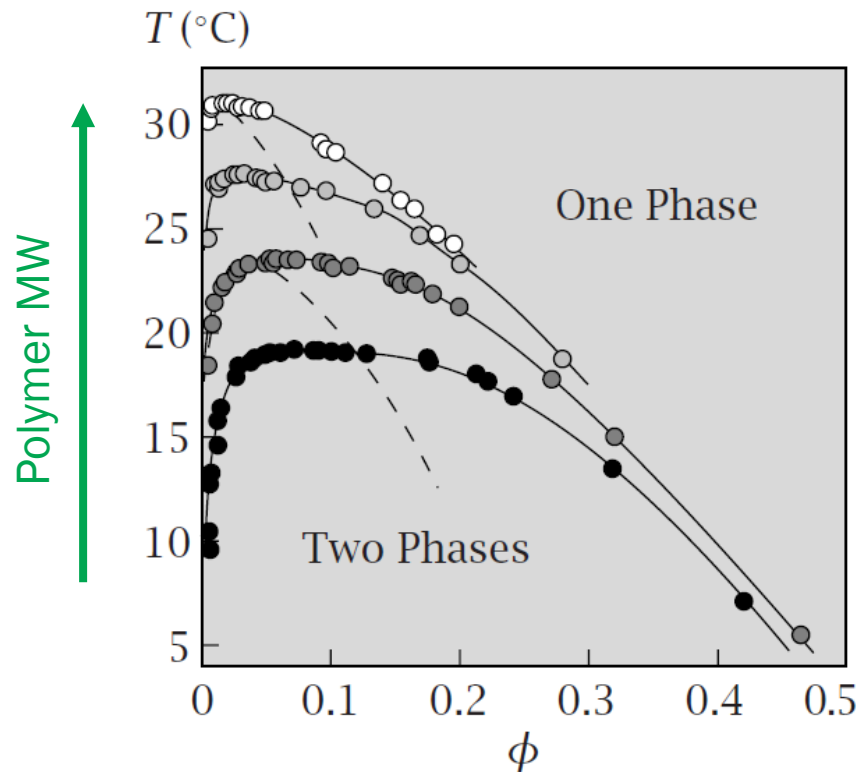


Phase Diagram



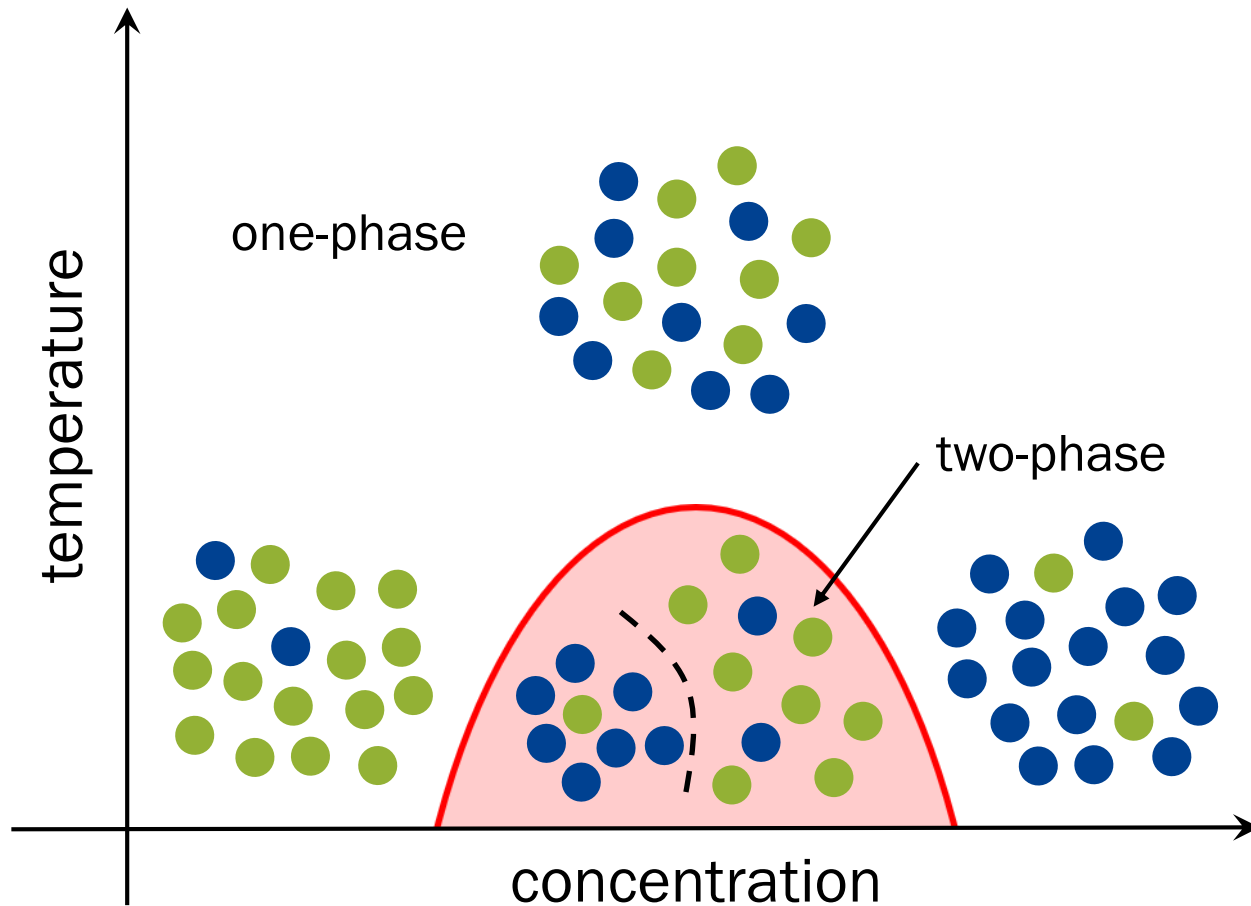
(Images: Dill and Bromberg, *Molecular Driving Forces*, 2nd Ed., Figures 32.7;
Brangwynne et al., *Nature Physics* (2015))

Polystyrene in Cyclohexane

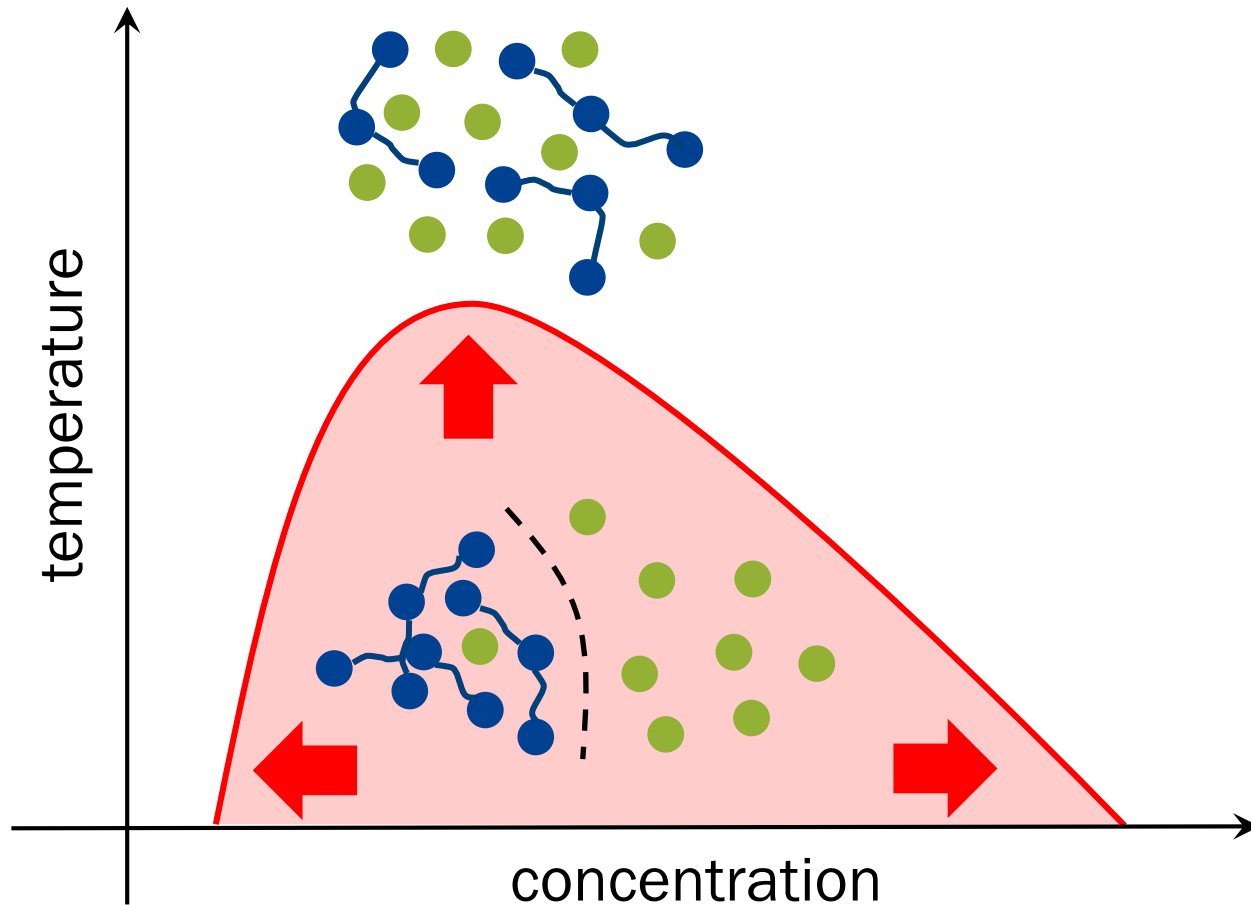


(Image: Dill and Bromberg, *Molecular Driving Forces*, 2nd Ed., Figures 32.8)

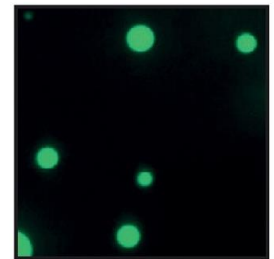
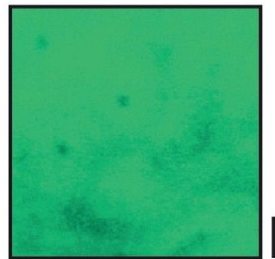
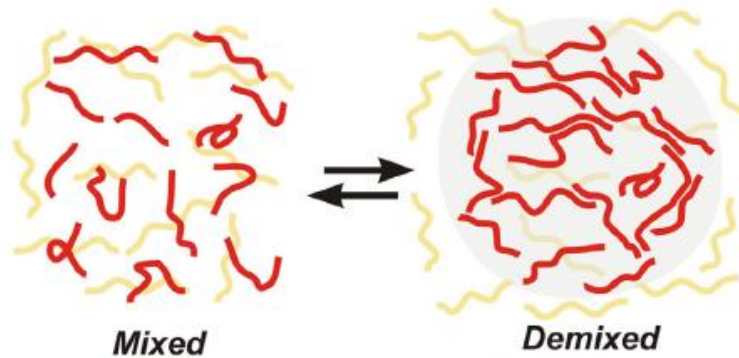
Phase Diagram



In Case of Polymer + Solvent

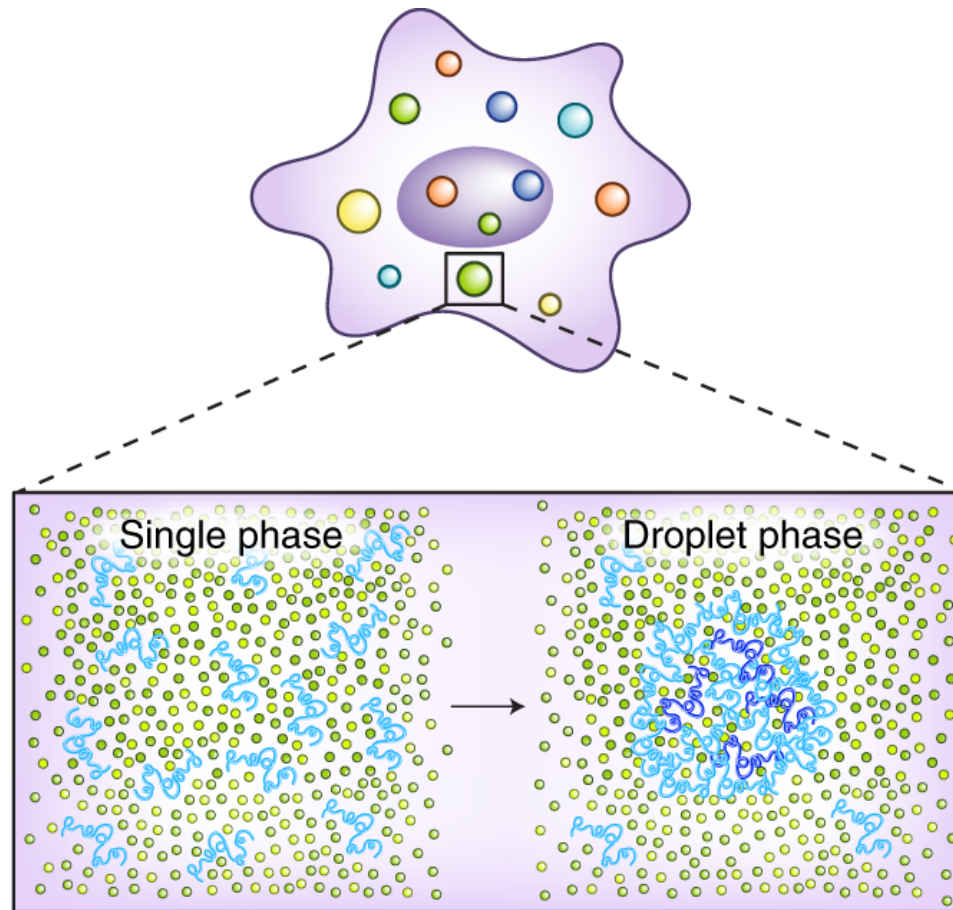


IDPs and Phase Separation

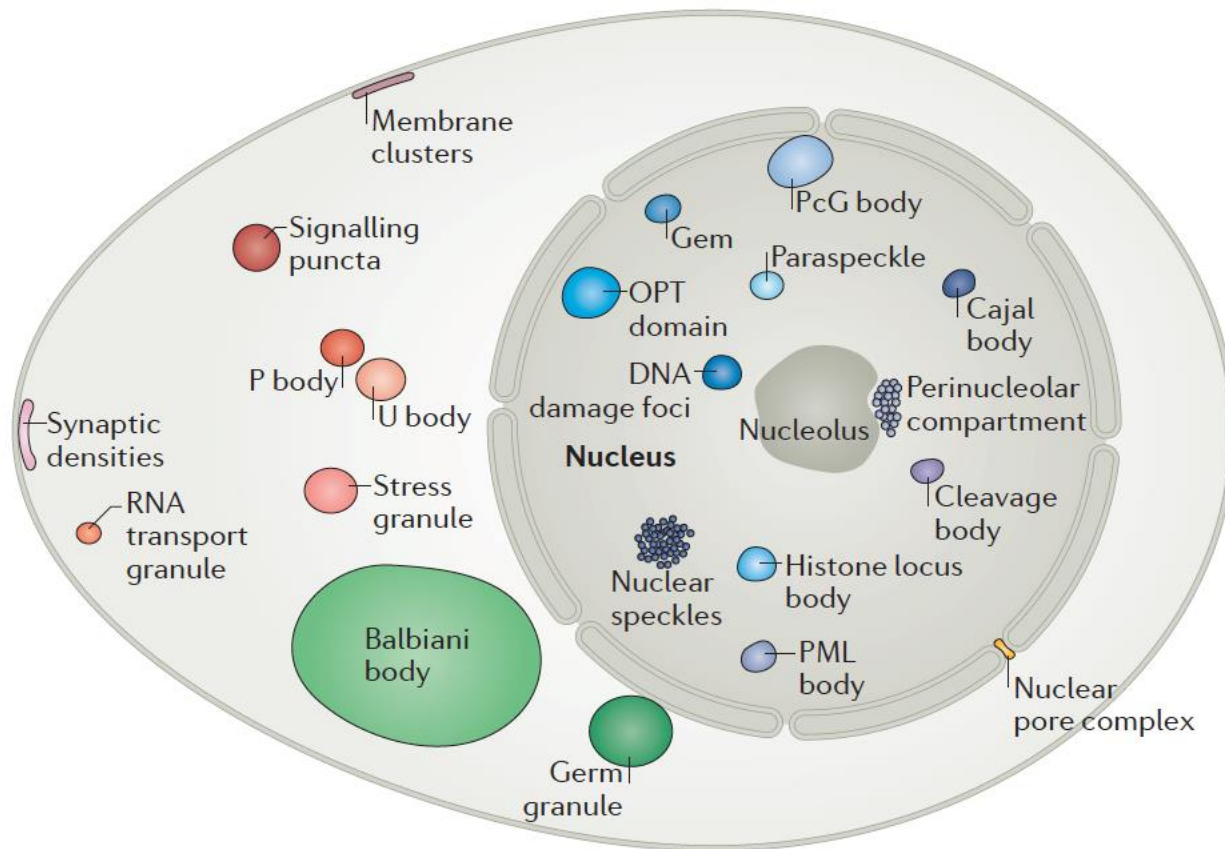


(Scale bars: 5 μm)

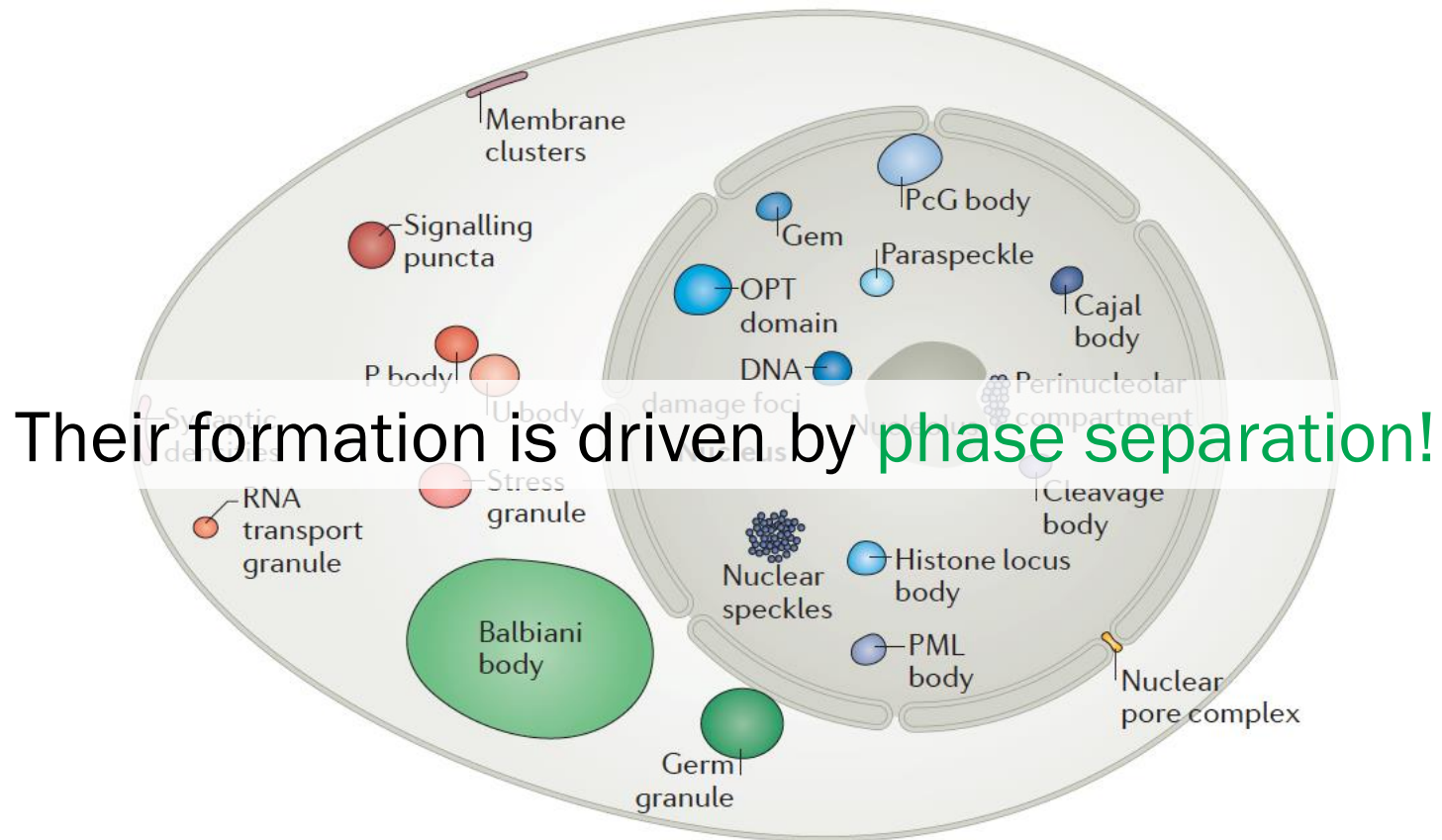
Intracellular Phase Separation



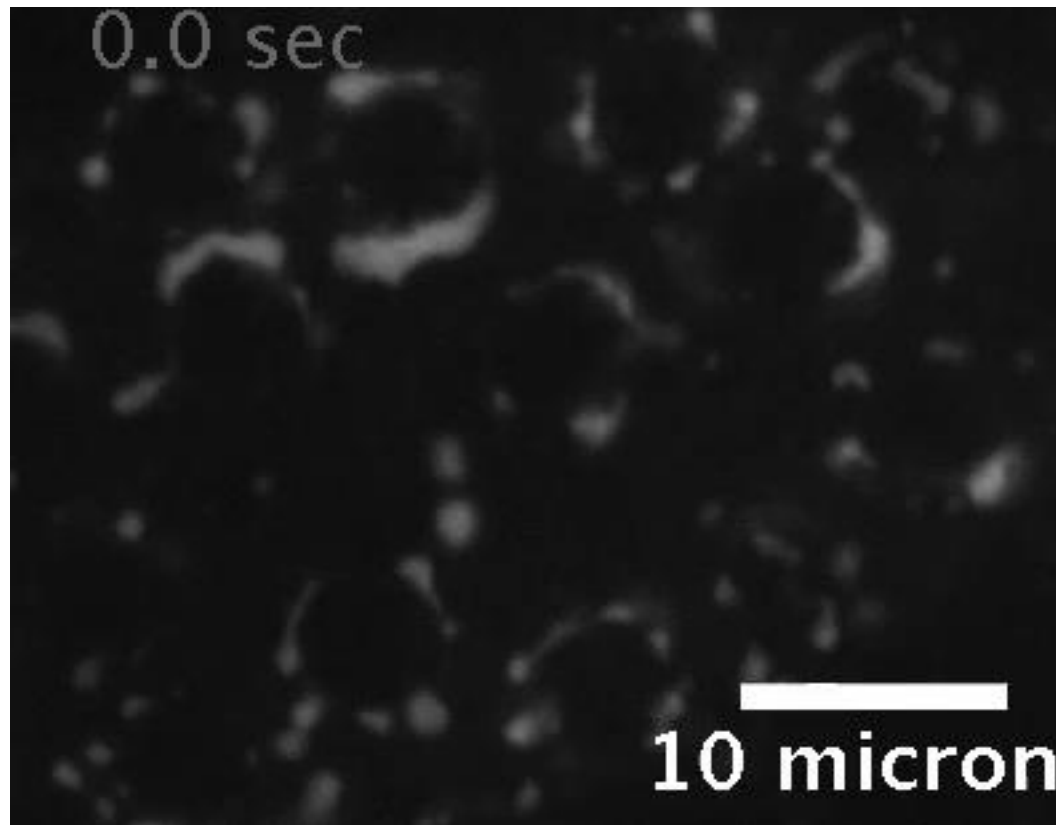
Membraneless Organelles (MLOs)



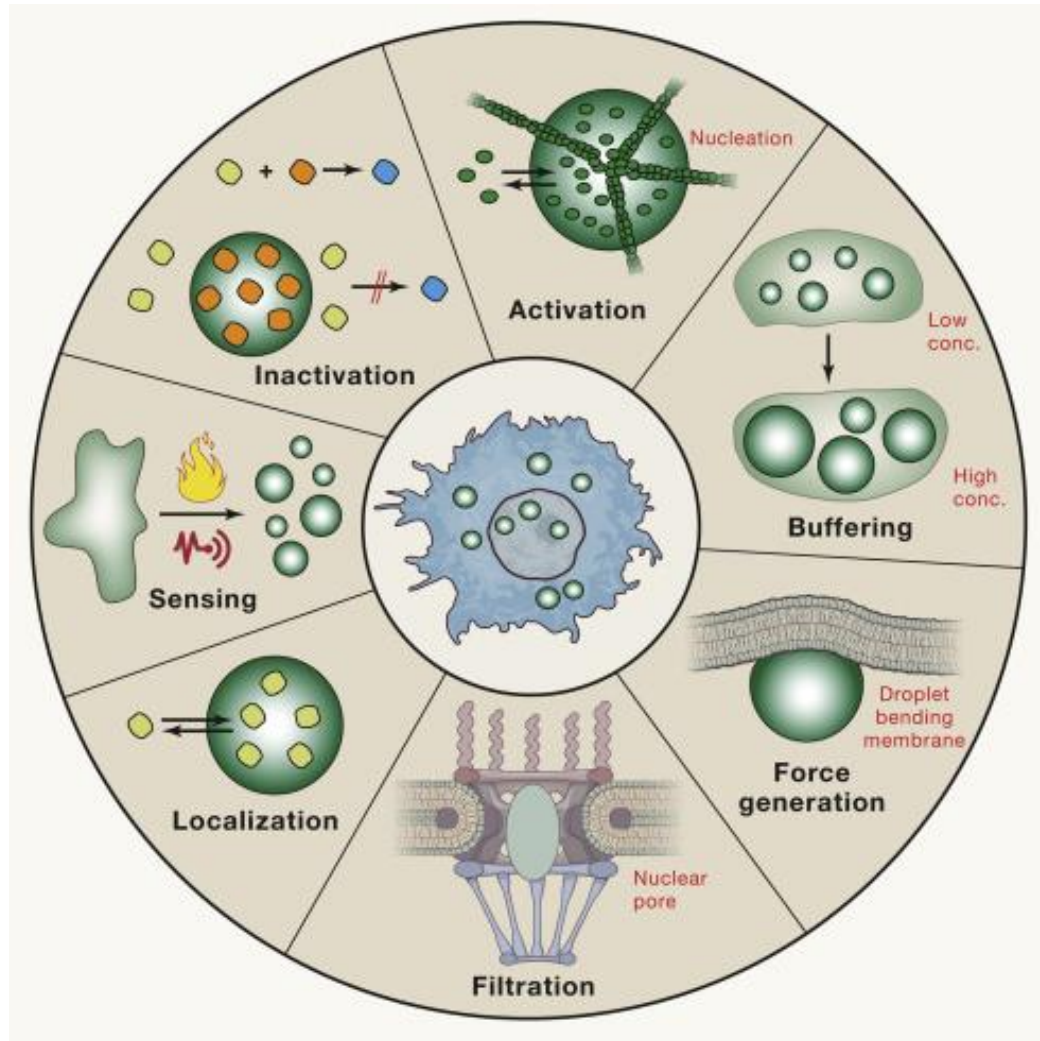
Membraneless Organelles (MLOs)



Liquidity of MLOs (even *in vivo*)

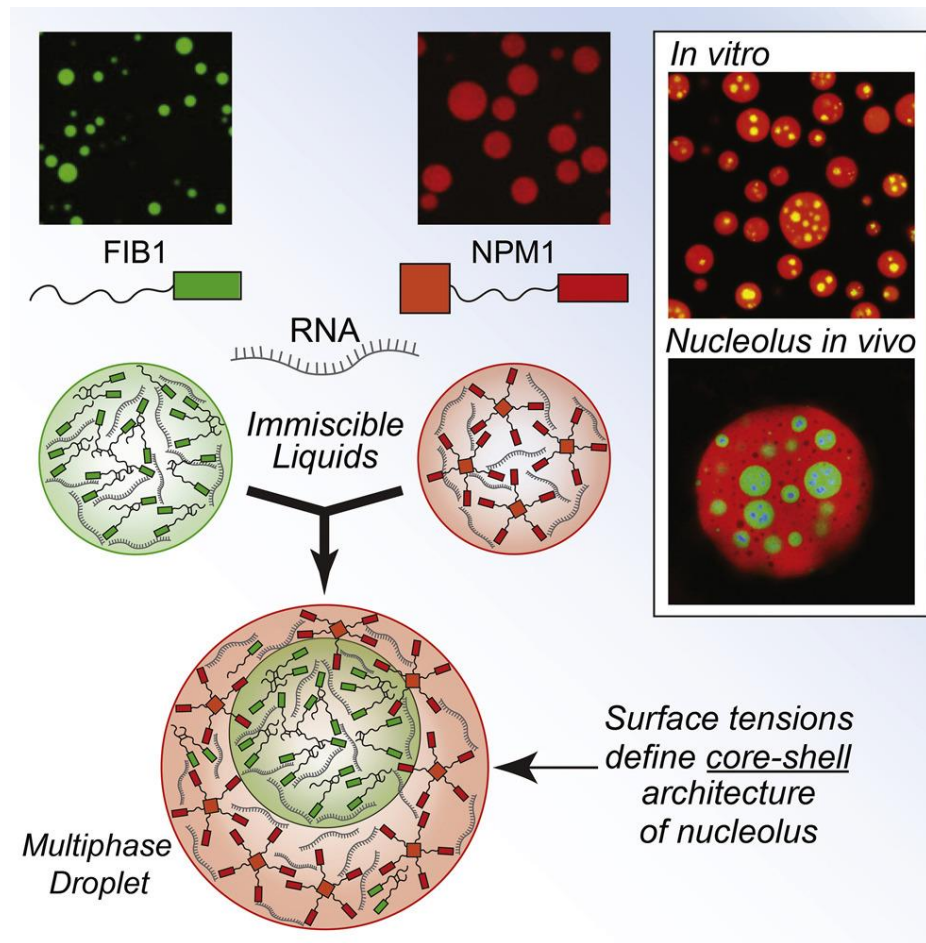


Biological Roles of MLOs

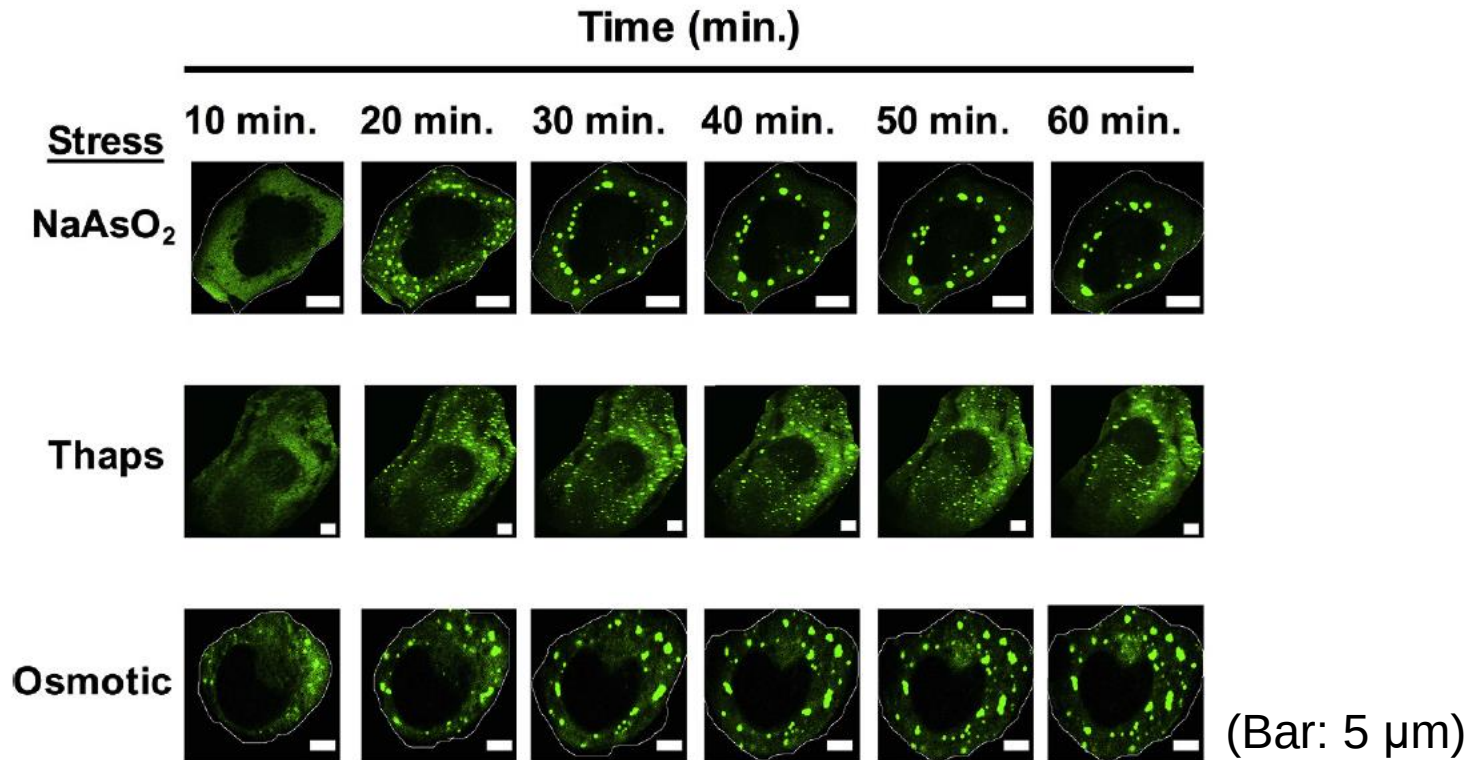


Alberti et al., *Cell* (2019)

Example: Nucleolus Formation



Example: Stress Granules



human osteosarcoma cells (U-2 OS)
Signal from GFP-G3BP1 fusion protein

Wheeler et al., *eLife* (2016)

Phase Separation in Neurons

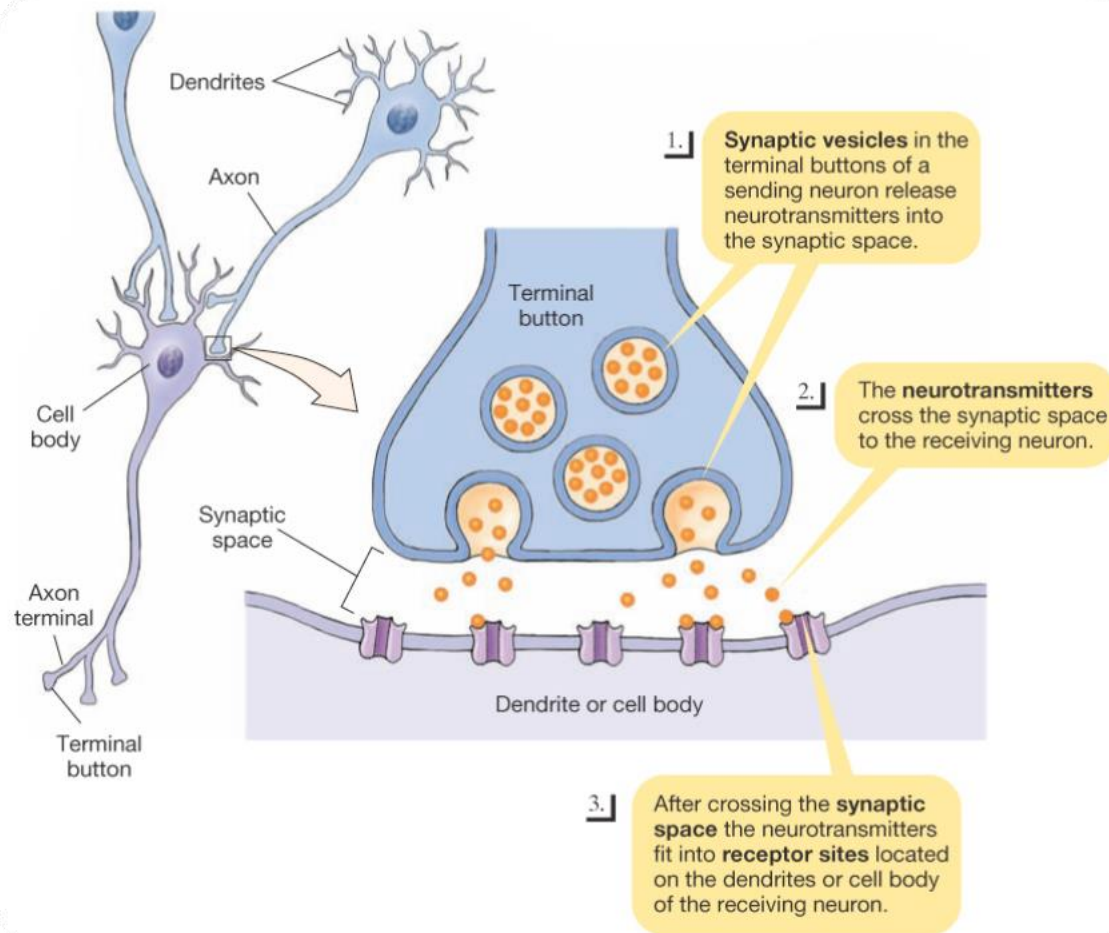
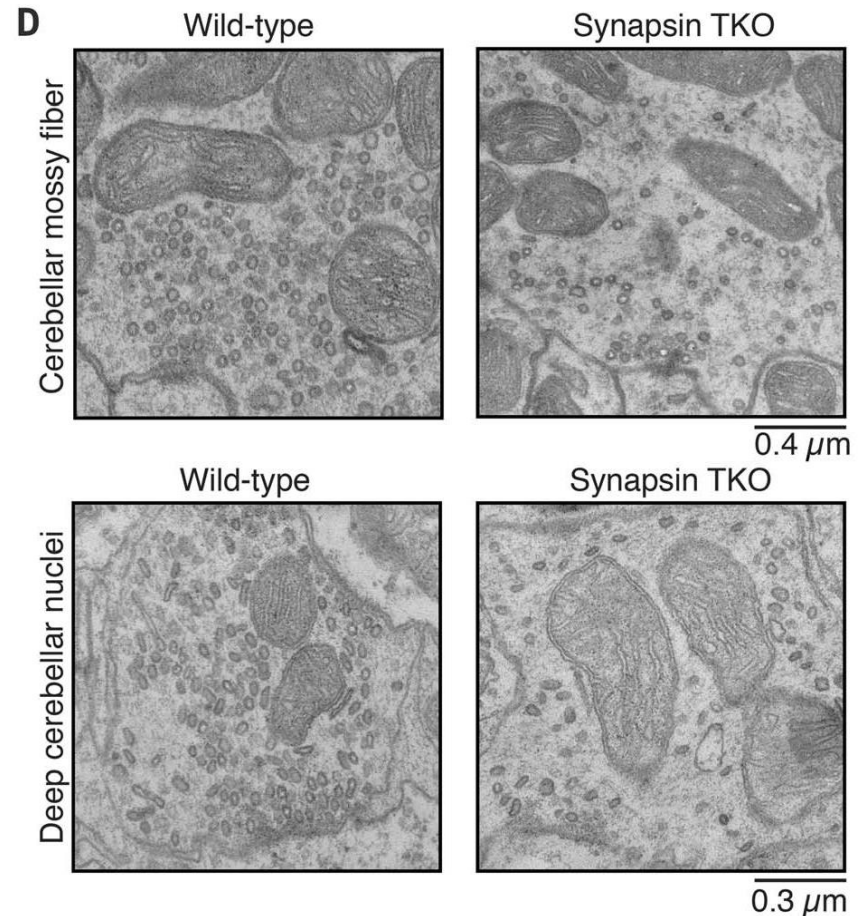
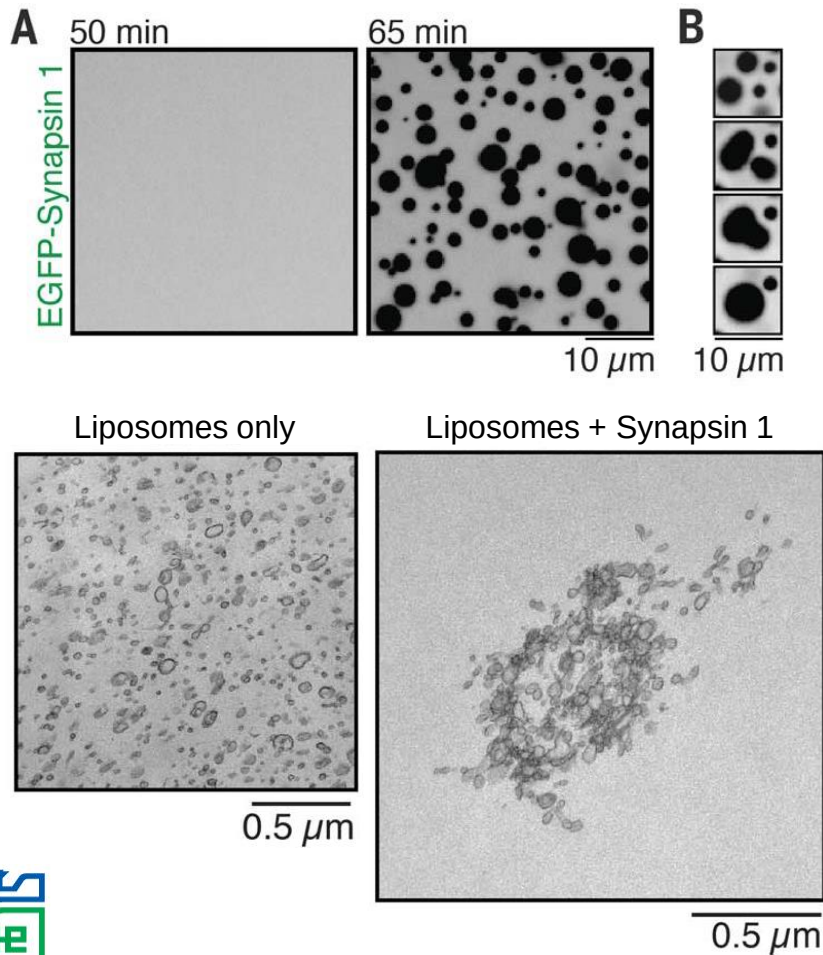


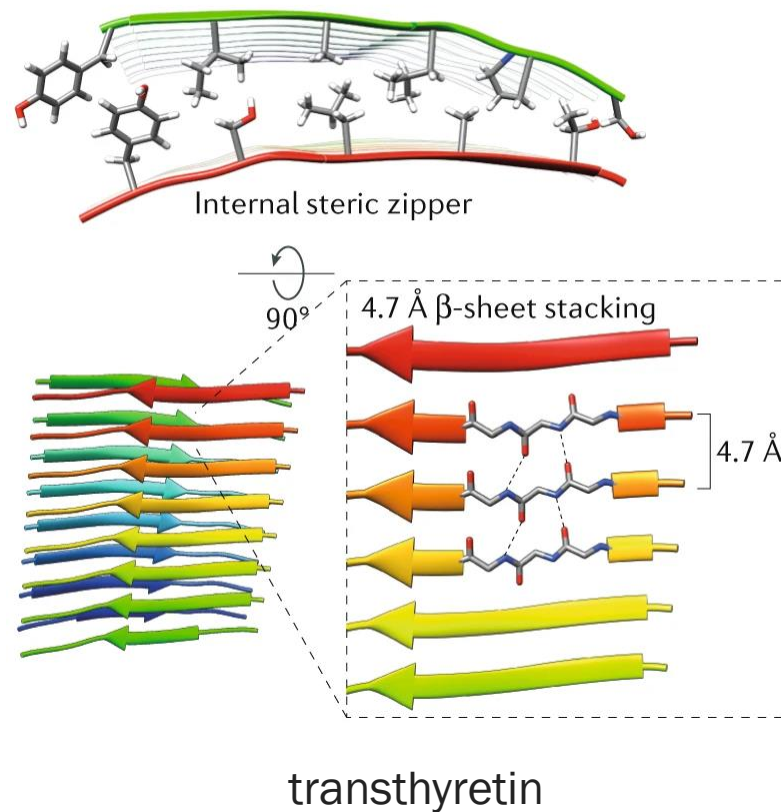
Image: <https://socratic.org/questions/how-do-impulses-travel-across-a-synapse>

Phase Separation in Neurons

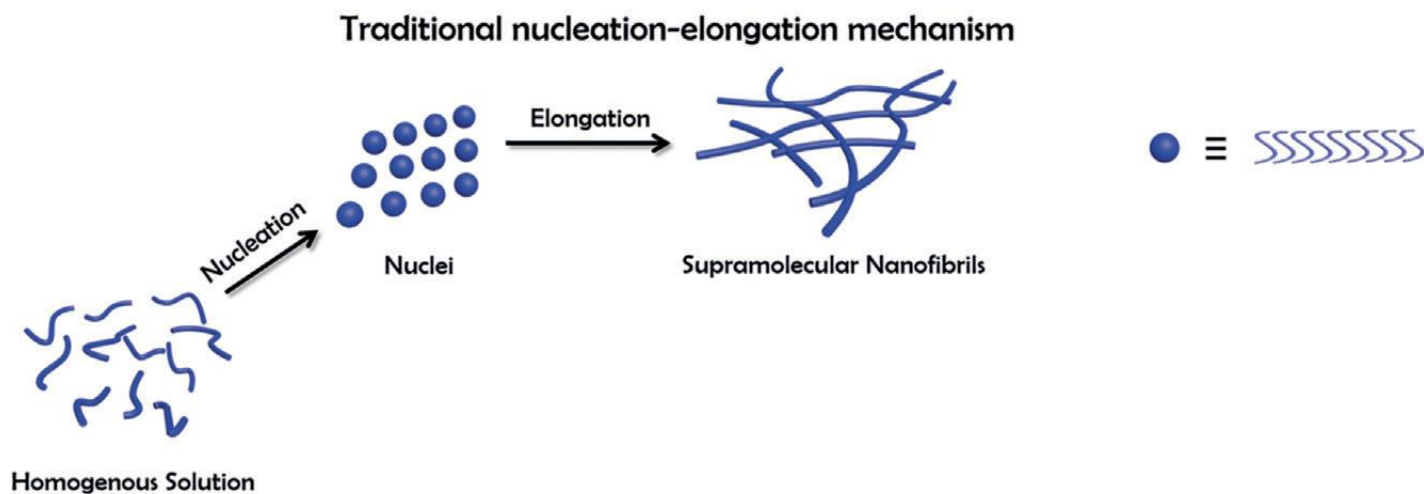


Milovanovic *et al.*, *Science* (2018)

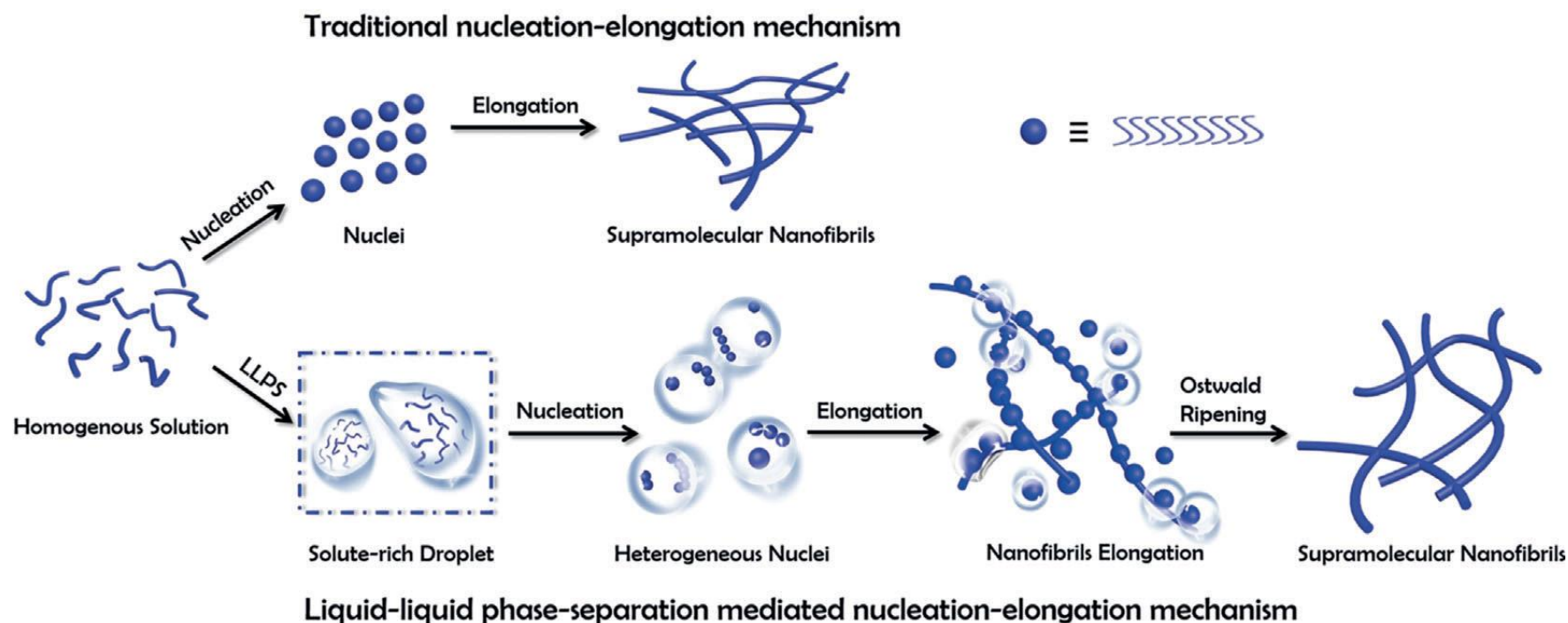
Fibril Formation Revisited



Traditional Hypothesis



Detour via Phase Separation



IDPs Show Rich Phase Behaviors

